REAL ANALYSIS II · SEMESTER 5 · MODULE 3

Answer key: The Riemann Integral

Formal solutions to all 40 questions of the Module 3 model paper (Bartle & Sherbert §7.1–§7.3).

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1 (Prepare). If one feels new, read that topic first, try writing the skeleton yourself, and then read the solution.

Grey "Marks" notes show a typical split (a study guide, not an official scheme): statement · setup · key step · conclusion.

Section A · Short answers

3 marks each. Ceiling 48.

1 PYQ MTS-26 Q4 also MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16

Definition of a partition and its norm; norms on [0,4] and [2,9].

KNOW FIRST

T01 M02

DEFINITION · PARTITION AND NORM (BARTLE §7.1, EQ. (1))

If I := [a,b] is a closed bounded interval in $\mathbb{R}$, a **partition** of I is a finite, ordered set $P := (x_0, x_1, \ldots, x_{n-1}, x_n)$ of points in I such that $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$. The **norm** (or mesh) of P is the length of the largest subinterval $I_i = [x_{i-1}, x_i]$: $\|P\| := \max\{x_1 - x_0,\ x_2 - x_1,\ \ldots,\ x_n - x_{n-1}\}$.

SKELETON

List $\Delta x_i = x_i - x_{i-1}$ for each partition; take $\|P\| = \max_i \Delta x_i$.

On [0,4]:

- $P_1 = (0, 1, 2, 4)$: subinterval lengths are 1-0=1, 2-1=1, 4-2=2, so $\|P_1\| = \max\{1, 1, 2\} = 2$.
- $P_2 = (0, 2, 3, 4)$: subinterval lengths are 2, 1, 1, so $\|P_2\| = \max\{2, 1, 1\} = 2$.
- $P_3 = (0, 1, 1.5, 2, 3.4, 4)$: subinterval lengths are 1, 0.5, 0.5, 1.4, 0.6, so $\|P_3\| = \max\{1, 0.5, 0.5, 1.4, 0.6\} = 1.4$.

On [2,9] (MAT-22 MCQ 16): $\|Q_1\| = \max\{2, 2, 2, 1\} = 2$; $\|Q_2\| = \max\{4, 3\} = 4$; $\|Q_3\| = \max\{3, 2, 2\} = 3$.

Thus $Q_2 = (2, 6, 9)$ has the largest norm (MCQ option B).

ANSWER $\|P_1\| = 2,\ \|P_2\| = 2,\ \|P_3\| = 1.4$**; largest on [2,9] is** Q_2 **with** $\|Q_2\| = 4$**.**

Marks: definitions 1 · norms on [0,4] 1 · largest norm on [2,9] 1.

2 PYQ MTS-22 Q4

Tagged partition and Riemann sum; $f(x) = x^2$ **on [0,4] with P = (0,1,2,4).**

KNOW FIRST

T02 P02 M03 Tags and endpoints: one point can tag two intervals

DEFINITION · TAGGED PARTITION AND RIEMANN SUM (BARTLE §7.1, EQ. (2))

If a point $t_i \in I_i = [x_{i-1}, x_i]$ is selected from each subinterval of a partition P for $i = 1, \ldots, n$, the points t_i are called **tags** and $\dot{P} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a **tagged partition** of [a,b]. For f : $[a, b] \to \mathbb{R}$, the **Riemann sum** of f corresponding to $\dot{P}$ is $S(f; \dot{P}) := \sum_{i=1}^n f(t_i)(x_i - x_{i-1})$.

SKELETON

Write the subintervals [0,1], [1,2], [2,4] with widths $\Delta x_1 = 1, \Delta x_2 = 1, \Delta x_3 = 2$, evaluate $f(t_i) = t_i^2$, and sum $f(t_1)\Delta x_1 + f(t_2)\Delta x_2 + f(t_3)\Delta x_3$.

Left-endpoint tags $t_1 = 0, t_2 = 1, t_3 = 2 : S(f; \dot{P}_{\text{left}}) = 0^2(1-0) + 1^2(2-1) + 2^2(4-2) = 0 + 1 + 8 = 9.$

Right-endpoint tags $t_1 = 1, t_2 = 2, t_3 = 4$: $S(f; \dot{P}_{\text{right}}) = 1^2(1-0) + 2^2(2-1) + 4^2(4-2) = 1 + 4 + 32 = 37.$

ANSWER $S(f; \dot{P}_{\text{left}}) = 9$ **and** $S(f; \dot{P}_{\text{right}}) = 37.$

Marks: definition 1 · left-endpoint sum 1 · right-endpoint sum 1.

3 PYQ MAT-21 MCQ 9 · adapted

Definition of $f \in \mathcal{R}[a,b]$; why $\int_a^b f$ is not an ordinary function limit.

KNOW FIRST

T03 M02 M03

DEFINITION 7.1.1 · RIEMANN INTEGRABLE (BARTLE P. 201)

A function f : $[a,b] \to \mathbb{R}$ is said to be **Riemann integrable** on [a,b] if there exists a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{P}$ is any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$, then $|S(f;\dot{P}) - L| < \varepsilon$. The set of all Riemann integrable functions on [a,b] is denoted by $\mathcal{R}[a,b]$, and we write $L = \int_a^b f$ or $\int_a^b f(x)\,dx$.

Why this is not an ordinary function limit as $\|P\| \to 0$ (Remark 7.1.1): For a given positive number $h = \|P\|$, there are infinitely many partitions P with norm h and infinitely many choices of tags $t_i \in I_i$ for each such P. Because many different Riemann sums $S(f;\dot{P})$ correspond to the same number $\|P\|$, $S(f;\dot{P})$ is not a single-valued function of $\|P\|$.(In MAT-21 MCQ 9, the function satisfying Definition 7.1.1 is called **Riemann integrable**, option A.)

Marks: Definition 7.1.1 word for word 1.5 · notation $\mathcal{R}[a,b]$ and $\int_a^b f 0.5$· non-function limit explanation 1.

4

Uniqueness of the Riemann integral (Theorem 7.1.2).

KNOW FIRST

T04 P05 Triangle inequality and the "arbitrary ε" rule M08

SKELETON

Assume L' and L'' both satisfy Definition 7.1.1. Given $\varepsilon > 0$, take $\delta'_{\varepsilon/2}$ and $\delta''_{\varepsilon/2}$, put $\delta_\varepsilon = \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\}$, and pick one $\dot{P}$ with $\|P\| < \delta_\varepsilon$ to get $|\,\text{L}' - L''| < \varepsilon$.

PROOF OF THEOREM 7.1.2 (BARTLE P. 201)

1. Suppose L' and L'' both satisfy Definition 7.1.1 for $f \in \mathcal{R}[a,b]$, and let $\varepsilon > 0$ be arbitrary.
   We will show $|\,\text{L}' - L''| < \varepsilon$ for every $\varepsilon > 0$ (P05).

2. By Definition 7.1.1 with tolerance $\varepsilon/2 > 0$, there exists $\delta'_{\varepsilon/2} > 0$ such that $\|P_1\| < \delta'_{\varepsilon/2} \Rightarrow |S(f;\dot{P}_1) - L'| < \varepsilon/2$, and there exists $\delta''_{\varepsilon/2} > 0$ such that $\|P_2\| < \delta''_{\varepsilon/2} \Rightarrow |S(f;\dot{P}_2) - L''| < \varepsilon/2$.
   Definition 7.1.1 applied to L' and L''.

3. Put $\delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0$, and choose a tagged partition $\dot{P}$ of [a,b] with $\|P\| < \delta_\varepsilon$. Then both inequalities of Step 2 hold for $\dot{P}$, so by the Triangle Inequality, $|\,\text{L}' - L''| \le |\,\text{L}' - S(f;\dot{P})| + |S(f;\dot{P}) - L''| < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, L' = L''. ∎
   Common δ_ε argument (M08), Triangle Inequality, and arbitrary ε rule (P05).

Marks: setup with ε/2 1 · δε = min\(δ'(ε/2),δ''(ε/2)\) and triangle inequality 1.5 · conclusion 0.5.

5 PYQ MTS-25 Q4

Constant function f(x) = k is in $\mathcal{R}[a,b]$ with $\int_a^b k = k(b-a)$.

KNOW FIRST

T05 T03 M01 Telescoping sums: why middle terms vanish

SKELETON

Compute $S(f;\dot{P}) = k(b-a)$ for every tagged partition $\dot{P}$; given $\varepsilon > 0$, take $\delta_\varepsilon = 1$.

PROOF (BARTLE EXAMPLE 7.1.4(A), P. 202)

1. Let f(x) := k for all $x \in [a,b]$. If $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is any tagged partition of [a,b], then $S(f;\dot{P}) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = \sum_{i=1}^n k(x_i - x_{i-1}) = k\sum_{i=1}^n (x_i - x_{i-1}) = k(x_n - x_0) = k(b-a)$.
   Factor out k (P02) and telescope $\sum_{i=1}^n (x_i - x_{i-1}) = b - a$ (M01).

2. Hence for any $\varepsilon > 0$, we can choose $\delta_\varepsilon := 1 > 0$ so that if $\dot{P}$ has $\|P\| < \delta_\varepsilon$, then $|S(f;\dot{P}) - k(b-a)| = |k(b-a) - k(b-a)| = 0 < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ and $\int_a^b k = k(b-a)$. ∎
   Definition 7.1.1 satisfied with L = k(b-a) and $\delta_\varepsilon = 1$.

ANSWER $\int_a^b k = k(b-a)$.

Marks: telescoping Riemann sum S(f;dot P) = k(b-a) 1.5 · explicit δε = 1 and | S - k(b-a)| = 0 < ε 1 · conclusion 0.5.

6 PYQ MAT-22 MCQ 18 · adapted | also MAT-21 MCQ 13, MTS-22 MCQ 7 · adapted

Changing $g \in \mathcal{R}[a,b]$ at finitely many points (Theorem $7.1.3$); $\int_0^2 h$; finite vs countable exception sets.

KNOW FIRST

T06 M03 M04

SKELETON

For one exceptional point c, bound $|S(f;\dot{P}) - S(g;\dot{P})| \leq 2(|f(c)| + |g(c)|)\|P\| < \varepsilon/2$ when $\|P\| < \delta_1$, and combine with $|S(g;\dot{P}) - L| < \varepsilon/2$.

PROOF OF THEOREM 7.1.3 (BARTLE PP. 201–202)

1. It suffices to prove the case of one exceptional point $c \in [a,b]$ (finitely many points follow by induction). Let $L := \int_a^b g$ and suppose f(x) = g(x) for all $x \neq c$.
   Induction reduces m exceptional points to 1 point.

2. For any tagged partition $\dot{P}$, the sums $S(f;\dot{P})$ and $S(g;\dot{P})$ agree except on at most two subintervals (when c is a partition point tag of two adjacent intervals). Thus $|S(f;\dot{P}) - S(g;\dot{P})| \leq 2(|f(c)| + |g(c)|)\|P\|$.
   At most 2 terms differ (M03), each of width $\leq \|P\| (M04)$.

3. Given $\varepsilon > 0$, choose $\delta_1 > 0$ with $\delta_1 < \varepsilon/(4(|f(c)| + |g(c)| + 1))$ and choose $\delta_2 > 0$ so that $\|P\| < \delta_2 \Rightarrow |S(g;\dot{P}) - L| < \varepsilon/2$. Let $\delta := \min\{\delta_1, \delta_2\} > 0$. For $\|P\| < \delta$, $|S(f;\dot{P}) - L| \leq |S(f;\dot{P}) - S(g;\dot{P})| + |S(g;\dot{P}) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$, so $f \in \mathcal{R}[a,b]$ and $\int_a^b f = L = \int_a^b g$. ∎
   Triangle Inequality and Definition 7.1.1.

Applications & MCQ resolutions:

- For h(x) = 2 on [0,1), h(1) = 3, h(x) = 1 on (1,2]: h differs from the step function $g = 2\varphi_{[0,1]} + 1\varphi_{(1,2]}$ only at c = 1, so $\int_0^2 h = \int_0^2 g = 2(1) + 1(1) = 3$.
- If f(x) = 0 except at finitely many points $c_1, \ldots, c_n (MAT-22$ MCQ 18, MAT-21 MCQ 13): comparing with $g \equiv 0$ gives **both** $f \in \mathcal{R}[a,b]$ (option A) **and** $\int_a^b f = 0$ (option B). (Note: in MAT-22 MCQ 18 both A and B are true; in MAT-21 MCQ 13 option B is the unique correct option.)

- **Ambiguous past question note (MTS-22 MCQ 7):** When f(x) = g(x) off the **countably infinite** set $\{1/n : n \in \mathbb{N}\}$, knowing $f \in \mathcal{R}[0,1]$ does **not** guarantee $g \in \mathcal{R}[0,1]$ in general (e.g. f = 0 and g(1/n) = n is unbounded, so $g \notin \mathcal{R}[0,1]$); if both $f, g \in \mathcal{R}[0,1]$ then $\int_0^1 f = \int_0^1 g$ since f g = 0 outside a null set. The intended exam key for MTS-22 MCQ 7 treats the unquantified statement (C) as the false option, or assumes boundedness.

ANSWER $\int_0^2 h = 3$**; if f=0 off a finite set,** $f \in \mathcal{R}[a,b]$ **and** $\int_a^b f = 0$**.**

Marks: proof of 7.1.3 1.5 · ∫02 h = 3 0.5 · finite vs countable discussion 1.

7

PYQ MAT-22 MCQ 15 · adapted

also MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9 · adapted

Boundedness Theorem 7.1.6; H(1/k) = k not integrable; converses.

KNOW FIRST

T08 P05

THEOREM 7.1.6 · BOUNDEDNESS THEOREM (BARTLE P. 205)

If $f \in \mathcal{R}[a,b]$, then f is bounded on [a,b].

SKELETON

Suppose $f \in \mathcal{R}[a,b]$ is unbounded. Fix $\delta > 0$ for $\varepsilon = 1$ so $|S(f;\dot{P})| < |L| + 1$. Fix a partition Q with $\|Q\| < \delta$; f is unbounded on some I_k; pick $t_k \in I_k$ huge to contradict $|S(f;\dot{Q})| < |L| + 1$.

PROOF OF THEOREM 7.1.6 (BARTLE PP. 205–206)

1. Suppose $f \in \mathcal{R}[a,b]$ with integral L, and assume for contradiction that f is unbounded on [a,b]. Taking $\varepsilon = 1$ in Definition 7.1.1 gives $\delta > 0$ such that $\|P\| < \delta \Rightarrow |S(f;\dot{P}) - L| < 1$, whence $|S(f;\dot{P})| < |L| + 1$.
   Definition 7.1.1 with $\varepsilon = 1$ and Triangle Inequality.

2. Let $Q = \{[x_{i-1}, x_i]\}_{i=1}^{n}$ be a partition of [a,b] with $\|Q\| < \delta$. Since f is unbounded on [a,b], it must be unbounded on at least one subinterval $[x_{k-1}, x_k]$.
   If f were bounded on each of the n pieces by M_i, it would be bounded on [a,b] by $\max_i M_i$.

3. Tag Q by setting $t_i := x_i$ for $i \neq k$, and choose $t_k \in [x_{k-1}, x_k]$ so that $|f(t_k)(x_k - x_{k-1})| > |L| + 1 + |\sum_{i \neq k} f(t_i)(x_i x_{i-1})|$. By the Reverse Triangle Inequality, $|S(f;\dot{Q})| \geq |f(t_k)(x_k - x_{k-1})| |\sum_{i \neq k} f(t_i)(x_i - x_{i-1})| > |L| + 1$, contradicting Step 1. Thus f is bounded. ∎
   Reverse Triangle Inequality (P05).

Applications & Converses:

- For $H(1/k) = k (k \in \mathbb{N}), H(x) = 0$ elsewhere on [0,1]: since $H(1/k) = k \to \infty$, H is unbounded on [0,1], so by Theorem $7.1.6, H \notin \mathcal{R}[0,1] (MTS - 22$ MCQ 9 option A; MAT-21 MCQ 17 option B; MTS-22 MCQ 8 option D).
- $f \in \mathcal{R}[a,b]$ does **not** imply f is continuous or monotone (Thomae's function C8 is integrable but neither continuous nor monotone; MAT-22 MCQ 15 and MAT-21 MCQ 11 option A).

- Not every bounded function is integrable: Dirichlet's function C7 is bounded $(0 \le f \le 1)$ but $f \notin \mathcal{R}[0,1]$.

Marks: statement & proof of 7.1.62· $H \notin \mathcal{R}[0,1]$ and converses 1.

8 PYQ MAT-21 MCQ 10 · adapted

Elementary step function φ_J (Lemma 7.2.4) and $\int_0^4 g$.

KNOW FIRST

T13 T06 M10

SKELETON

For J = [c,d] and $\|P\| < \delta$, trap $S(\varphi_{[c,d]}; \dot{P})$ between $(d-c) - 2\delta$ and $(d-c) + 2\delta$ and pick $\delta_\varepsilon = \varepsilon/4$; extend to (c,d), [c,d), (c,d] via Theorem 7.1.3.

PROOF OF LEMMA 7.2.4 (BARTLE P. 210)

1. First let $J = [c,d] \subseteq [a,b] (c < d)$. For any tagged partition $\dot{P}$ with $\|P\| < \delta$, the union U_1 of subintervals in $\dot{P}$ with tag $t_i \in [c,d]$ satisfies $[c+\delta, d-\delta] \subseteq U_1 \subseteq [c-\delta, d+\delta] (M10)$, so $(d-c) - 2\delta \leq S(\varphi_{[c,d]}; \dot{P}) \leq (d-c) + 2\delta$.
   Tag-inclusion bounds (M10 / Exercise 7.1.13).

2. Given $\varepsilon > 0$, take $\delta_\varepsilon := \varepsilon/4 > 0$; then $\|P\| < \delta_\varepsilon \Rightarrow |S(\varphi_{[c,d]}; \dot{P}) - (d-c)| \leq 2(\varepsilon/4) = \varepsilon/2 < \varepsilon$. Thus $\varphi_{[c,d]} \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_{[c,d]} = d - c$.
   Definition 7.1.1 with $\delta_\varepsilon = \varepsilon/4$.

3. The other three intervals (c,d), [c,d), (c,d] have indicator functions differing from $\varphi_{[c,d]}$ at at most the two endpoints c, d. By Theorem 7.1.3, each $\varphi_J \in \mathcal{R}[a,b]$ with $\int_a^b \varphi_J = d - c$. ∎
   Theorem 7.1.3 (T06).

Evaluation of $\int_0^4 g (MAT-21$ MCQ 10): Since g(x) = 3 on [0,2] and g(x) = 2 on (2,4], we write $g = 3\varphi_{[0,2]} + 2\varphi_{(2,4]}$. By Lemma 7.2.4 and Theorem 7.1.5, $\int_0^4 g = 3(2-0) + 2(4-2) = 6 + 4 = 10 (MCQ$ option C).

ANSWER $\int_a^b \varphi_J = d - c$ **and** $\int_0^4 g = 10.$

Marks: proof for [c,d] 1.5 · extension by $7.1.30.5 \cdot \int_0^4 g = 101.$

9

Oriented integrals (Definition 7.2.12, Theorem 7.2.13) and check for $f(x) = x, \alpha = 3, \beta = 1, \gamma = 0$.

KNOW FIRST

T19 T17

DEFINITION 7.2.12 & THEOREM 7.2.13 (BARTLE P. 214)

If $f \in \mathcal{R}[a,b]$ and $\alpha, \beta \in [a,b]$ with $\alpha < \beta$, we define $\int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0$. **Theorem 7.2.13:** For any numbers $\alpha, \beta, \gamma \in [a,b]$, $\displaystyle\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f$.

Verification for f(x) = x with $\alpha = 3, \beta = 1, \gamma = 0$:

Using $\int_u^v x\,dx = \frac{1}{2}(v^2 - u^2)$ for u < v:

- $\displaystyle\int_3^1 x\,dx = -\int_1^3 x\,dx = -\tfrac{1}{2}(3^2 - 1^2) =$ -4.
- $\displaystyle\int_3^0 x\,dx + \int_0^1 x\,dx = -\int_0^3 x\,dx + \int_0^1 x\,dx = -\tfrac{1}{2}(9-0) + \tfrac{1}{2}(1-0) = -\frac{9}{2} + \frac{1}{2} =$ -4.

ANSWER **Both sides equal -4.**

Marks: Definition 7.2.12 1 · Theorem 7.2.13 statement 0.5 · verification 1.5.

10

Why FTC 1 fails for $H(x) = 2\sqrt{x}$ on [0,b] and $K(x) = x^2\cos(1/x^2)$ on [0,1].

KNOW FIRST

T21 T20 Fundamental Theorem of Calculus (First Form)

T08

Theorem 7.3.1 (FTC 1) requires three hypotheses: (a) F continuous on [a,b]; (b) F'(x) = f(x) on $[a,b] \setminus E$ for a finite set E; and $(c) f \in \mathcal{R}[a,b]$. In both examples, **hypothesis (c) fails** because the derivative is unbounded (Theorem 7.1.6):

- **For $H(x) = 2\sqrt{x}$ on [0,b] (Example 7.3.2(d)):** H is continuous on [0,b] and $H'(x) = \dfrac{1}{\sqrt{x}}$ on $(0,b) (E = \{0\})$. However, $h(x) = \dfrac{1}{\sqrt{x}}$ is **unbounded** on (0,b) as $x \to 0^+$, so by Theorem $7.1.6, h \notin \mathcal{R}[0,b]$ regardless of how h(0) is defined.
- **For $K(x) = x^2 \cos(1/x^2) (x \in (0,1]), K(0) = 0$ on [0,1] (Example 7.3.2(e), C19):** For x > 0, the Product and Chain Rules give $K'(x) = 2x\cos(1/x^2) + \dfrac{2}{x}\sin(1/x^2)$, and at $x = 0, K'(0) = \lim_{x \to 0} \dfrac{x^2 \cos(1/x^2)}{x} = 0$. Thus K is continuous and differentiable at **every** point of $[0,1] (E = \varnothing)$. Yet along $x_m = (\dfrac{\pi}{2} + 2m\pi)^{-1/2} \to 0^+, \sin(1/x_m^2) = 1$ and $K'(x_m) = \dfrac{2}{x_m} \to \infty$, so K' is unbounded on [0,1] and K' $\notin \mathcal{R}[0,1]$.

Marks: identify hypothesis (c) f∈$\mathscr{R}$ via 7.1.6 1 · H'(x)=1/√ x unbounded 1 · K'(x) formula, K'(0)=0, and unboundedness 1.

11 PYQ MAT-21 Q6

Definition of the indefinite integral $F(z) = \int_a^z f$; proof that F is Lipschitz and continuous (Theorem 7.3.4).

KNOW FIRST

T22 T17 T07

DEFINITION 7.3.3 · INDEFINITE INTEGRAL (BARTLE P. 218)

If $f \in \mathcal{R}[a,b]$, the function defined by $F(z) := \int_a^z f \quad \text{for } z \in [a,b]$ is called the **indefinite integral** of f with **basepoint** a.

PROOF OF THEOREM 7.3.4 (BARTLE P. 218)

1. Let $w, z \in [a,b]$ with $w \le z$. By the Additivity Theorem 7.2.9, $F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f$, so $F(z) - F(w) = \int_w^z f$.
   Additivity Theorem 7.2.9 (T17).

2. Since $|f(x)| \le M$ on [a,b], we have -M $\le f(x) \le M$ on [w,z]. By Theorem 7.1.5(c) and Example $7.1.4(a)$, $-M(z-w) \le \int_w^z f \le M(z-w)$, whence $|F(z) - F(w)| = |\int_w^z f| \le M|z-w|$ for all $z, w \in [a,b]$.
   Monotonicity of the integral (7.1.5c) and constant integral (7.1.4a).

3. Thus F is Lipschitz on [a,b] with constant M, and hence uniformly continuous and continuous on [a,b] (given $\varepsilon > 0$, take $\delta := \varepsilon/(M+1)$). ∎
   Lipschitz ⇒ continuous (P09).

Marks: Definition 7.3.3 1 · Additivity step 1 · Lipschitz bound and continuity 1.

12

Indefinite integrals of $\operatorname{sgn} x$ on [-1,1] and Thomae's h on [0,1] (Examples 7.3.7(a),(b)).

KNOW FIRST

T24 T23 T09

1. For $f(x) = \operatorname{sgn} x$ on [-1,1] with basepoint -1 (Example 7.3.7(a), C20):

- If -1 $\leq x <$ 0, then f(t) = -1 on [-1,x), so $F(x) = \int_{-1}^{x}(-1)\,dt = -(x+1) = |x| - 1$.
- If $0 \leq x \leq 1$, Additivity gives $F(x) = \int_{-1}^{0}(-1)\,dt + \int_{0}^{x} 1\,dt =$ -1 $+x = |x| - 1$.

Thus $F(x) = |x| - 1$ for all $x \in [-1, 1]$. Since $F'_{-}(0) =$ -1 $\neq +1 = F'_{+}(0)$, $F'(0)$ does not exist, so F is **not** an antiderivative of f on [-1,1].

2. For Thomae's function h on [0,1] (Example 7.3.7(b), C8): Since $\int_0^1 h = 0$ and $h(x) \geq 0$, we have $0 \leq H(x) = \int_0^x h \leq \int_0^1 h = 0$, so H(x) = 0 for all $x \in [0, 1]$. Although H'(x) = 0 exists everywhere on [0,1], at every rational $x = m/n \in \mathbb{Q} \cap [0, 1]$ we have $H'(x) = 0 \neq 1/n = h(x)$, so H is **not** an antiderivative of h on [0,1].

ANSWER **$F(x) = |x| - 1 (F'(0) DNE); H(x) = 0 (H'(x) \neq h(x)$ on $\mathbb{Q} \cap [0, 1])$. Neither is an antiderivative.**

Marks: F(x) = | x| - 1 on [-1,1] 1 · F'(0) DNE 1 · Thomae H equiv 0 and H' ≠ h on rationals 1.

13 PYQ MTS-25 Q7 also MTS-26 Q7, MTS-22 MCQ 10

Definition of a null set; proof that $\mathbb{Q} \cap [0,1]$ is null; properties of null sets.

KNOW FIRST

T26 P03 P13 Countable sets and the Archimedean property M07

DEFINITION 7.3.10(A) · NULL SET (BARTLE P. 220)

A set $Z \subseteq \mathbb{R}$ is said to be a **null set** if for every $\varepsilon > 0$ there exists a countable collection $\{(a_k, b_k)\}_{k=1}^{\infty}$ of open intervals such that $Z \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k)$ and $\sum_{k=1}^{\infty}(b_k a_k) \leq \varepsilon$.

PROOF THAT $\mathbb{Q}_1 = \mathbb{Q} \cap [0,1]$ IS A NULL SET (EXAMPLE 7.3.11, P. 221)

1. Since $\mathbb{Q}_1 = \mathbb{Q} \cap [0,1]$ is countable, enumerate its elements as $\mathbb{Q}_1 = \{r_1, r_2, \ldots, r_k, \ldots\}$.
   Countability of $\mathbb{Q} \cap [0,1] (P13)$.

2. Given $\varepsilon > 0$, for each $k \in \mathbb{N}$ define the open interval $J_k := (r_k - \frac{\varepsilon}{2^{k+1}},\ r_k + \frac{\varepsilon}{2^{k+1}})$, which contains r_k and has length $\frac{\varepsilon}{2^k}$.
   Geometric halving trick (M07).

3. Then $\mathbb{Q}_1 \subseteq \bigcup_{k=1}^{\infty} J_k$ and the total length is $\sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon \sum_{k=1}^{\infty} \frac{1}{2^k} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\mathbb{Q} \cap [0,1]$ is a null set. ∎
   Geometric series $\sum_{k=1}^{\infty} 1/2^k = 1 (P03)$.

Properties (MTS-22 MCQ 10): Any subset of a null set is null (A true); the union of two null sets is null (B true); every countable set is null by the same proof (C true). However, **not every null set is countable** (D is the **false** statement): the Cantor set is an uncountable null set.

Marks: Definition 7.3.10(a) 1 · proof for $\mathbb{Q}\cap[0,1]$ 1.5 · properties & Cantor set remark 0.5.

14 PYQ MAT-22 MCQ 27, 28, 29 · adapted | also MAT-21 MCQ 19, 20 · adapted

FTC 1 evaluations: $\int_a^b f = F(b)-F(a)$, $\int_{-2}^{3} g = 5/2$, and $\int_a^b |x|\,dx = B(b)-B(a)$.

KNOW FIRST

T20 Fundamental Theorem of Calculus (First Form) T21

(a) MAT-22 MCQ 27 / MAT-21 MCQ 19: If f is continuous on [a,b] (hence $f \in \mathcal{R}[a,b]$ by 7.2.7) and F is any antiderivative of f on [a,b] (F is continuous by 6.1.2 and F' = f with $E=\varnothing$), then by Theorem 7.3.1, $\int_a^b f(x)\,dx = F(b)-F(a) (MCQ$ option B).

(b) Bartle Ex 7.3.3 (MAT-22 MCQ 28 / MAT-21 MCQ 20): On $[-2,3]$, $G(x)=\frac{1}{2}|x^2-1|$ is continuous. For $|x|>1$, $G(x)=\frac{1}{2}(x^2-1)$ has G'(x) = x = g(x); for $|x|<1$, $G(x)=\frac{1}{2}(1-x^2)$ has G'(x) = -x = g(x). Thus G'(x) = g(x) off the finite set $E=\{-1,1\}$, and $g\in\mathcal{R}[-2,3]$ (piecewise continuous). By FTC 1, $\int_{-2}^{3} g(x)\,dx = G(3)-G(-2) = \frac{1}{2}|9-1|-\frac{1}{2}|4-1| = 4-\frac{3}{2} = \frac{5}{2}$ (MCQ option B).

(c) Bartle Ex 7.3.4 (MAT-22 MCQ 29): For $B(x) = -\frac{1}{2}x^2 (x<0), \frac{1}{2}x^2 (x\ge 0)$, we have B'(x) = -x = $|x|$ for x < 0, $B'(x)=x=|x|$ for x > 0, and $B'_-(0)=0=B'_+(0)=|0|$. Thus B is continuous and $B'(x)=|x|$ on all of [a,b] with $E=\varnothing$, and $|x|\in\mathcal{R}[a,b]$ (continuous). By FTC 1, $\int_a^b |x|\,dx = B(b)-B(a) (MCQ$ option B).

ANSWER $(a) F(b)-F(a); (b)\int_{-2}^{3} g = \frac{5}{2} (E=\{-1,1\}); (c)\int_a^b |x|\,dx = B(b)-B(a) (E=\varnothing)$.

Marks: part (a) 0.5 · part (b) with E=\(-1,1\) and value 5/2 1.5 · part (c) with E=varnothing 1.

15 PYQ MAT-22 Q35

Evaluate $\int_0^1 x^2(1-\sqrt{x})\,dx$ using the Fundamental Theorem of Calculus.

KNOW FIRST

T20 T21 T15

SKELETON

Write $f(x) = x^2 - x^{5/2}$ and $F(x) = \frac{1}{3}x^3 - \frac{2}{7}x^{7/2}$ on [0,1]. Check hypotheses (a), (b), (c) of Theorem 7.3.1 with $E = \varnothing$, then compute F(1) - F(0).

SOLUTION

1. On [0,1], expand the integrand as $f(x) := x^2(1-\sqrt{x}) = x^2 - x^{5/2}$. Since f is continuous on the closed bounded interval [0,1], Theorem 7.2.7 implies that $f \in \mathcal{R}[0,1]$ (hypothesis (c) of Theorem 7.3.1).
   Continuous ⇒ integrable (Theorem 7.2.7, T15).

2. Define F : $[0,1] \to \mathbb{R}$ by $F(x) := \dfrac{x^3}{3} - \dfrac{2}{7}x^{7/2}$. Then F is continuous on [0,1] (hypothesis (a)), and $F'(x) = x^2 - x^{5/2} = f(x)$ for all $x \in [0,1]$ (hypothesis (b) with $E = \varnothing$).
   Power rule for derivatives on [0,1].

3. By the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1), $\int_0^1 x^2(1-\sqrt{x})\,dx = F(1) - F(0) = (\frac{1}{3} - \frac{2}{7}) - 0 = \dfrac{7-6}{21} = \dfrac{1}{21}$. ■
   Theorem 7.3.1 (T20).

ANSWER $\displaystyle\int_0^1 x^2(1-\sqrt{x})\,dx = \frac{1}{21}.$

Marks: $f \in \mathcal{R}[0,1]$ by continuity 1 · antiderivative $F(x) = \frac{1}{3}x^3\frac{2}{7}x^{7/2}1\cdot F(1)-F(0) = 1/211.$

16 PYQ MAT-22 MCQ 19, 26 · adapted

Trigonometric symmetry $\int_0^{\pi/2} f(\cos x)\,dx$ and translation $\int_{a+c}^{b+c} f(y-c)\,dy = \int_a^b f$.

KNOW FIRST

T25 T19

(a) MAT-22 MCQ 26: Let $\varphi(x) := \frac{\pi}{2} - x$ on $[0, \pi/2]$. Since $\varphi'(x) =$ -1 is continuous and $\cos x = \sin(\frac{\pi}{2} - x)$, Theorem 7.3.8 and Definition 7.2.12 give $\int_0^{\pi/2} f(\cos x)\,dx = -\int_0^{\pi/2} f(\sin(\varphi(x)))\,\varphi'(x)\,dx = -\int_{\pi/2}^0 f(\sin u)\,du = \int_0^{\pi/2} f(\sin u)\,du$. Next, on $[\pi/2, \pi]$, substituting $u = \psi(x) := \pi - x (\psi' =$ -1, $\sin x = \sin(\pi - x))$ gives $\int_{\pi/2}^{\pi} f(\sin x)\,dx = \int_0^{\pi/2} f(\sin u)\,du$. By Additivity 7.2.9, $\int_0^{\pi} f(\sin x)\,dx = 2\int_0^{\pi/2} f(\sin x)\,dx$, so $\int_0^{\pi/2} f(\cos x)\,dx = \frac{1}{2}\int_0^{\pi} f(\sin x)\,dx$. Thus options (A) and (B) are equal to $\int_0^{\pi/2} f(\cos x)\,dx$, while $(C)\ \int_0^{\pi} f(\sin x)\,dx$ **is not equal to it** (twice as large whenever the integral is non-zero).

(b) Bartle Ex 7.1.15 / MAT-22 MCQ 19: For g(y) = f(y - c) on [a+c,b+c], any tagged partition $\dot{P}_c = \{([y_{i-1}, y_i], s_i)\}$ of [a+c,b+c] shifts by -c to a tagged partition $\dot{P} = \{([y_{i-1} - c, y_i - c], s_i - c)\}$ of [a,b] with the same norm and $S(g; \dot{P}_c) = S(f; \dot{P})$. Thus $g \in \mathcal{R}[a+c, b+c]$ and $\int_{a+c}^{b+c} g = \int_a^b f (MCQ$ option B; when f is continuous this also follows from Theorem 7.3.8 with $\varphi(y) = y - c, \varphi'(y) = 1$).

ANSWER **(a) Not equal to** $(C)\ \int_0^{\pi} f(\sin x)\,dx$**;** $(b)\ \int_{a+c}^{b+c} g = \int_a^b f$ **(option B).**

Marks: ∫0(pi/2) f(cos x)\,dx = ∫0(pi/2) f(sin x)\,dx 1 · tfrac12∫0pi f(sin x)\,dx 1 · translate formula 1.

17 PYQ MAT-22 MCQ 30 · adapted

If f : $[0,1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0,1]$, then $f \equiv 0$.

KNOW FIRST

T23 T22 T17 T15

SKELETON

Set $F(x) = \int_0^x f$; use Additivity to write $\int_x^1 f = F(1) - F(x)$, deduce $F(x) = \frac{1}{2}F(1)$ is constant, and differentiate via Theorem 7.3.6.

PROOF (BARTLE EXERCISE 7.3.16)

1. Since f is continuous on [0,1], Theorem 7.2.7 implies that $f \in \mathcal{R}[0,1]$. Define the indefinite integral $F(x) := \int_0^x f$ for $x \in [0,1]$.
   Theorem 7.2.7 (T15) proves $(B) f \in \mathcal{R}[0,1]$.

2. By the Additivity Theorem 7.2.9, $F(1) = \int_0^1 f = \int_0^x f + \int_x^1 f$. Since $\int_0^x f = \int_x^1 f$ by hypothesis, we have 2F(x) = F(1), so $F(x) = \frac{1}{2}F(1)$ is constant on [0,1] (in fact, setting x = 0 gives F(1) = 2F(0) = 0, so F(x) = 0 for all $x \in [0,1]$).
   Additivity Theorem 7.2.9 (T17).

3. Since f is continuous on [0,1], Theorem 7.3.6 (FTC 2) gives f(x) = F'(x) = 0 for all $x \in [0,1]$. ∎
   Theorem 7.3.6 (T23). (In MAT-22 MCQ 30, both $(A) f(x) = 0\ \forall x$ and $(B) f \in \mathcal{R}[0,1]$ hold, so the answer is (C) "All the above".)

Marks: f ∈ $\mathscr{R}$[0,1] 0.5 · 2F(x) = F(1) ⇒ F equiv 0 via Additivity 1.5 · f = F' = 0 via FTC 2 1.

18 PYQ MAT-22 MCQ 23 · adapted

If f is continuous on $[a,b]$, $f \geq 0$, and $\int_a^b f = 0$, then f(x) = 0 for all $x \in [a,b]$.

KNOW FIRST

T15 T13 T07

SKELETON

Suppose f(c) > 0 for some $c \in [a,b]$. Use continuity at c with $\varepsilon = \frac{1}{2}f(c)$ to find a closed subinterval $J \subseteq [a,b]$ of length d-c > 0 where $f \geq \frac{1}{2}f(c)$, getting $\int_a^b f \geq \frac{1}{2}f(c)(d-c) > 0$.

PROOF (BARTLE EXERCISES 7.2.8–7.2.9)

1. Suppose for contradiction that there is some $c \in [a,b]$ with f(c) > 0. Since f is continuous at c, taking $\varepsilon := \frac{1}{2}f(c) > 0$ gives $\delta > 0$ such that if $x \in [a,b]$ and $|x-c| < \delta$, then $f(x) > f(c) - \frac{1}{2}f(c) = \frac{1}{2}f(c)$.
   Definition of continuity at c with $\varepsilon = \frac{1}{2}f(c)$.

2. Choose a non-degenerate closed subinterval $J = [c_1, d_1] \subseteq [a,b] \cap (c-\delta, c+\delta)$ with $c_1 < d_1$. Since $f(x) \geq 0$ on [a,b] and $f(x) > \frac{1}{2}f(c)$ on J, we have $\frac{1}{2}f(c)\varphi_J(x) \leq f(x)$ for all $x \in [a,b]$.
   Lower bound by an elementary step function (M06).

3. By Lemma 7.2.4 and Theorem 7.1.5(c), $\int_a^b f \geq \int_a^b \frac{1}{2}f(c)\varphi_J = \frac{1}{2}f(c)(d_1 - c_1) > 0$, which contradicts $\int_a^b f = 0$. Therefore f(x) = 0 for all $x \in [a,b]$ ($MAT-22$ MCQ 23 option A). ∎
   Monotonicity 7.1.5(c) and Lemma 7.2.4 (T13).

Why continuity cannot be dropped (Ex 7.2.9): Define f(a) := 1 and f(x) := 0 for $x \in (a, b]$. Then $f(x) \geq 0$ on [a,b] and $\int_a^b f = 0$ by Theorem 7.1.3, yet $f(a) = 1 \neq 0$.

Marks: continuity neighborhood with f > tfrac12 f(c) 1 · step-function lower bound ∫ab f > 0 1 · counterexample without continuity 1.

19 PYQ MTS-22 Q14

Leibniz rule for variable-limit integrals; derivatives of $\int_0^{x^2} (1+t^3)^{-1}\,dt$ and $\int_{x^2}^{x} \sqrt{1+t^2}\,dt$.

KNOW FIRST

T23 T19 P11

Leibniz rule for variable limits (Bartle Ex 7.3.10; PYQ MTS-22 Q14): Let f : $[a, b] \to \mathbb{R}$ be continuous and let u, v : $[c, d] \to [a, b]$ be differentiable. If $I(z) := \int_a^z f(t)\,dt$, then I'(z) = f(z) by Theorem 7.3.6 (FTC 2). Since $G(x) = \int_{u(x)}^{v(x)} f(t)\,dt = I(v(x)) - I(u(x))$ by Theorem 7.2.13, the Chain Rule 6.1.6 gives $G'(x) = I'(v(x))\,v'(x) - I'(u(x))\,u'(x) = f(v(x))\,v'(x) - f(u(x))\,u'(x)$.

$(a) F(x) = \int_0^{x^2} (1+t^3)^{-1}\,dt$ **(Bartle Ex 7.3.11(a)):** Here $f(t) = (1+t^3)^{-1}$ is continuous on [0,1], u(x) = 0, and $v(x) = x^2 (v'(x) = 2x)$, so $F'(x) = \dfrac{1}{1+(x^2)^3} \cdot 2x = \dfrac{2x}{1+x^6}$.

$(b) F(x) = \int_{x^2}^{x} \sqrt{1+t^2}\,dt$ **(Bartle Ex 7.3.11(b)):** Here $f(t) = \sqrt{1+t^2}$ is continuous on [0,1], v(x) = x (v'=1), and $u(x) = x^2 (u' = 2x)$, so $F'(x) = \sqrt{1+x^2} \cdot 1 - \sqrt{1+(x^2)^2} \cdot 2x = \sqrt{1+x^2} - 2x\sqrt{1+x^4}$.

ANSWER $(a) F'(x) = \dfrac{2x}{1+x^6}$; $(b) F'(x) = \sqrt{1+x^2} - 2x\sqrt{1+x^4}$.

Marks: general rule via FTC 2 + Chain Rule 1 · part (a) 1 · part (b) 1.

20

Mean Value Theorem for Integrals (Bartle Exercise 7.2.16).

KNOW FIRST

T15 T07 P07

PROOF (BARTLE EXERCISE 7.2.16)

1. Since f is continuous on the closed bounded interval $[a,b]$, $f \in \mathcal{R}[a,b]$ (Theorem 7.2.7) and by the Maximum–Minimum Theorem 5.3.4 (P07) f attains its minimum $m := \min_{[a,b]} f$ and maximum $M := \max_{[a,b]} f$ on [a,b].
   Theorem 7.2.7 and Max–Min Theorem 5.3.4 (P07).

2. Since $m \le f(x) \le M$ for all $x \in [a,b]$, Theorem 7.1.5(c) and Example 7.1.4(a) give $m(b-a) \le \int_a^b f \le M(b-a)$, so $m \le \dfrac{1}{b-a}\int_a^b f \le M$.
   Monotonicity of the integral (7.1.5c) and division by b - a > 0.

3. By Bolzano's Intermediate Value Theorem 5.3.7 (P07), f attains every value between m and M, so there exists $c \in [a,b]$ such that $f(c) = \dfrac{1}{b-a}\int_a^b f$, i.e. $\int_a^b f = f(c)(b-a)$. ∎
   Bolzano's IVT 5.3.7. (Alternative: apply the MVT 6.2.4 to $F(x) = \int_a^x f$, which gives $c \in (a,b)$.)

Fails for step functions: On [0,2], g(x) = 1 on [0,1) and g(x) = 2 on [1,2] has $\int_0^2 g = 3$, but $g(c)(2-0) = 3 \iff g(c) = 3/2$, which g never attains.

Marks: m(b-a) ≤ ∫ab f ≤ M(b-a) 1 · IVT conclusion 1 · step-function counterexample 1.

Section B · Paragraph / problem answers (Part 1: Q21–Q28)

6 marks each. Ceiling 72.

21

PYQ MAT-21 Q29

also MTS-22 Q5, MTS-25 Q5, MAT-22 MCQ 14

Linearity $\int_a^b (f+g) = \int_a^b f + \int_a^b g$ and monotonicity $f \le g \Rightarrow \int_a^b f \le \int_a^b g$ (Theorem 7.1.5(b),(c)).

KNOW FIRST

T07 T03 P02 P05 Triangle inequality and the "arbitrary ε" rule M08

THEOREM 7.1.5(B), (C) (BARTLE P. 204)

Suppose f and g are in $\mathcal{R}[a,b]$. Then:
(b) The function f + g is in $\mathcal{R}[a,b]$ and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

(c) If $f(x) \le g(x)$ for all $x \in [a,b]$, then $\int_a^b f \le \int_a^b g$.

SKELETON

Check $S(f+g;\dot{P}) = S(f;\dot{P}) + S(g;\dot{P})$ and $f \le g \Rightarrow S(f;\dot{P}) \le S(g;\dot{P})$. Given $\varepsilon > 0$, choose $\delta_\varepsilon = \min\{\delta_1, \delta_2\}$ so both sums are within $\varepsilon/2$ of their integrals. Use Triangle Inequality for (b) and sandwiching for (c).

PROOF OF THEOREM 7.1.5(B), (C) (BARTLE PP. 204–205)

1. For any tagged partition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ of [a,b], properties of finite sums (P02) yield $S(f+g;\dot{P}) = \sum_{i=1}^n (f(t_i) + g(t_i))(x_i - x_{i-1}) = S(f;\dot{P}) + S(g;\dot{P})$, and if $f(x) \leq g(x)$ on [a,b], then since $x_i - x_{i-1} > 0$ we have $f(t_i)(x_i - x_{i-1}) \leq g(t_i)(x_i x_{i-1})$, so $S(f;\dot{P}) \leq S(g;\dot{P})$.
   Additivity and order preservation of finite sums (P02).

2. Let $\varepsilon > 0$ be given. Since $f, g \in \mathcal{R}[a,b]$, Definition 7.1.1 provides $\delta_1 > 0$ and $\delta_2 > 0$ such that $\|P\| < \delta_1 \implies |S(f;\dot{P}) - \int_a^b f| < \frac{\varepsilon}{2}$, $\|P\| < \delta_2 \implies |S(g;\dot{P}) - \int_a^b g| < \frac{\varepsilon}{2}$. Define $\delta_\varepsilon := \min\{\delta_1, \delta_2\} > 0$, and let $\dot{P}$ be any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$.
   Definition 7.1.1 with tolerance $\varepsilon/2$ and common $\delta_\varepsilon(M08)$.

3. **Proof of (b):** By Step 1, Step 2, and the Triangle Inequality, $|S(f+g;\dot{P}) - (\int_a^b f + \int_a^b g)| = |(S(f;\dot{P}) \int_a^b f) + (S(g;\dot{P}) - \int_a^b g)| \leq |S(f;\dot{P}) - \int_a^b f| + |S(g;\dot{P}) - \int_a^b g| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f + g \in \mathcal{R}[a,b]$ and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.
   Triangle Inequality (P05) and Definition 7.1.1.

4 **Proof of (c):** For the same $\dot{P}$ with $\|P\| < \delta_\varepsilon$, unpacking the two absolute value inequalities of Step 2 gives $\int_a^b f - \frac{\varepsilon}{2} < S(f;\dot{P}) \quad \text{and} \quad S(g;\dot{P}) < \int_a^b g + \frac{\varepsilon}{2}$. Combining these with $S(f;\dot{P}) \le S(g;\dot{P})$ from Step 1 gives $\int_a^b f \frac{\varepsilon}{2} < \int_a^b g + \frac{\varepsilon}{2}$, or $\int_a^b f < \int_a^b g + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\int_a^b f \le \int_a^b g$. ∎

Arbitrary ε rule (P05).

Why strict inequality < need not hold (MAT-22 MCQ 14): Option $(B)\ \int_a^b f \le \int_a^b g$ is the true statement; strict < fails whenever f = g, or when $f \equiv 0$ and g(a) = 1, g(x) = 0 on (a,b] (where $\int_a^b f = 0 = \int_a^b g$).

Marks: Riemann sum relations 1 · common δε setup 1 · proof of (b) 2 · proof of (c) and MCQ remark 2.

22

Thomae's function h on [0,1]: definition, proof that $h \in \mathcal{R}[0,1]$ with $\int_0^1 h = 0$, and the set E for $h(x) \geq 1/5$.

KNOW FIRST

T09 T03 P14 M03 Tags and endpoints: one point can tag two intervals M10

DEFINITION OF THOMAE'S FUNCTION (BARTLE EXAMPLE 7.1.7, P. 206)

Define h : $[0,1] \to \mathbb{R}$ by h(x) := 0 if $x \in [0,1]$ is irrational, h(0) := 1, and h(x) := 1/n if $x \in [0,1]$ is the rational number x = m/n where $m, n \in \mathbb{N}$ have no common integer factor except 1.

SKELETON

Given $\varepsilon > 0$, let $E = \{x \in [0,1] : h(x) \geq \varepsilon/2\}$, which has finitely many points n. Choose $\delta = \varepsilon/(4n)$. Split $\dot{P}$ into $\dot{P}_1$ (tags in E, at most 2n subintervals) and $\dot{P}_2$ (tags outside E, where h < $\varepsilon/2$).

PROOF THAT $h \in \mathcal{R}[0,1]$ AND $\int_0^1 h = 0$ (BARTLE P. 206)

1. Let $\varepsilon > 0$ be given. Since $h(m/n) = 1/n \geq \varepsilon/2 \iff n \leq 2/\varepsilon$, there are only finitely many denominators $n \in \mathbb{N}$ and finitely many numerators $m \in \{1, \ldots, n\}$ satisfying $1/n \geq \varepsilon/2$. Thus the set $E := \{x \in [0,1] : h(x) \geq \varepsilon/2\}$ is a finite set.
   Archimedean property (P13).

2. Let n be the number of points in E, and define $\delta := \dfrac{\varepsilon}{4n} > 0$. Let $\dot{P}$ be any tagged partition of [0,1] with $\|P\| < \delta$. Separate $\dot{P}$ into $\dot{P}_1 (the$ subset of subintervals with tags in E) and $\dot{P}_2 (the$ subset with tags elsewhere in [0,1]).
   Tag-splitting tool (M10).

3. Allowing for a point of E serving as an endpoint tag of two adjacent subintervals, $\dot{P}_1$ has at most 2n subintervals, of total length at most $2n\delta = 2n \cdot \dfrac{\varepsilon}{4n} = \dfrac{\varepsilon}{2}$. Since $0 < h(t_i) \leq 1$ on $\dot{P}_1$, we have $S(h; \dot{P}_1) \leq 1 \cdot 2n\delta = \dfrac{\varepsilon}{2}$.
   Endpoint tag count (M03) and $h(x) \leq 1$.

4. For tags t_i in $\dot{P}_2$, we have $t_i \notin E$, so $0 \leq h(t_i) < \varepsilon/2$. Since the total length of the subintervals in $\dot{P}_2$ is at most 1 (the length of [0,1]), we have $S(h; \dot{P}_2) < \dfrac{\varepsilon}{2} \cdot 1 = \dfrac{\varepsilon}{2}$.
   Height $< \varepsilon/2$ times total width $\leq 1 (M04)$.

5. Adding Steps 3 and 4 gives $0 \leq S(h; \dot{P}) = S(h; \dot{P}_1) + S(h; \dot{P}_2) < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $h \in \mathcal{R}[0,1]$ and $\int_0^1 h = 0$. ∎
   Definition 7.1.1 with L = 0.

Explicit list of $E = \{x \in [0,1] : h(x) \geq 1/5\}$: We collect 0 (where h(0)=1) and all fractions $m/n \in (0,1]$ in lowest terms with denominator $n \in \{1,2,3,4,5\}$:

ANSWER $E = \{0,\ 1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{2}{3},\ \frac{1}{4},\ \frac{3}{4},\ \frac{1}{5},\ \frac{2}{5},\ \frac{3}{5},\ \frac{4}{5}\}$ (11 points); $\int_0^1 h = 0$.

Marks: definition of h 1 · finiteness of E and δ = ε/(4n) 1.5 · estimates on dot P1 and dot P2 2.5 · list of 11 points in E 1.

23 PYQ MAT-21 Q8

State and prove the Cauchy Criterion for Riemann integrability (Theorem 7.2.1).

KNOW FIRST

T10 T03 P06

THEOREM 7.2.1 · CAUCHY CRITERION (BARTLE P. 208)

A function f : $[a,b] \to \mathbb{R}$ belongs to $\mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $\dot{P}$ and $\dot{Q}$ are any tagged partitions of [a,b] with $\|P\| < \eta_\varepsilon$ and $\|Q\| < \eta_\varepsilon$, then $|S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon$.

SKELETON

($\Rightarrow$) Put $\eta_\varepsilon = \delta_{\varepsilon/2}$ and use the Triangle Inequality. ($\Leftarrow$) Take decreasing $\delta_n > 0$ for $\varepsilon = 1/n$, choose $\dot{P}_n$ with $\|P_n\| < \delta_n$, show $(S(f;\dot{P}_m))$ is Cauchy in $\mathbb{R}$ with limit A, let $m \to \infty$ to get $|S(f;\dot{P}_n) - A| \leq 1/n$, and pick $K > 2/\varepsilon$.

PROOF OF THEOREM 7.2.1 (BARTLE P. 208)

($\Rightarrow$)

1. Suppose $f \in \mathcal{R}[a,b]$ with integral L. Given $\varepsilon > 0$, let $\eta_\varepsilon := \delta_{\varepsilon/2} > 0$ from Definition 7.1.1. If $\dot{P}, \dot{Q}$ have $\|P\| < \eta_\varepsilon$ and $\|Q\| < \eta_\varepsilon$, then $|S(f;\dot{P}) - L| < \varepsilon/2$ and $|S(f;\dot{Q}) - L| < \varepsilon/2$, so $|S(f;\dot{P}) - S(f;\dot{Q})| \leq |S(f;\dot{P}) - L| + |L - S(f;\dot{Q})| < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$.
   Definition 7.1.1 and Triangle Inequality (P05).

($\Leftarrow$)

1. For each $n \in \mathbb{N}$, let $\delta_n > 0$ be the number $\eta_{1/n}$ such that $\|P\|, \|Q\| < \delta_n \Rightarrow |S(f;\dot{P}) - S(f;\dot{Q})| < 1/n$. By replacing δ_n with $\min\{\delta_1, \ldots, \delta_n\}$ if necessary, we may assume $\delta_n \geq \delta_{n+1}$ for all $n \in \mathbb{N}$.
   Hypothesis with $\varepsilon = 1/n$; forcing δ_n decreasing ensures $\|P_m\| < \delta_m \leq \delta_n$ for m > n.

2. For each $n \in \mathbb{N}$, choose a tagged partition $\dot{P}_n$ with $\|P_n\| < \delta_n$. If m > n, then both $\dot{P}_m$ and $\dot{P}_n$ have norm $< \delta_n$, so $|S(f;\dot{P}_m) - S(f;\dot{P}_n)| < \dfrac{1}{n}$.
   Step 1 applied to $\dot{P}_m$ and $\dot{P}_n$.

3. Thus $(S(f;\dot{P}_m))_{m=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}$. By Theorem 3.5.5 (P06), it converges to a real number $A := \lim_m S(f;\dot{P}_m)$. Letting $m \to \infty$ in Step 2 gives $|S(f;\dot{P}_n) - A| \leq 1/n$ for all $n \in \mathbb{N}$.
   Cauchy Convergence Criterion for sequences in $\mathbb{R}(P06)$.

4 Given $\varepsilon > 0$, choose $K \in \mathbb{N}$ with $K > 2/\varepsilon$. For any tagged partition $\dot{Q}$ with $\|Q\| < \delta_K$, $|S(f;\dot{Q}) - A| \le |S(f;\dot{Q}) - S(f;\dot{P}_K)| + |S(f;\dot{P}_K) - A| < \dfrac{1}{K} + \dfrac{1}{K} = \dfrac{2}{K} < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ with $\int_a^b f = A$. ∎

Triangle Inequality and Definition 7.1.1 with $\delta_\varepsilon = \delta_K$.

Marks: statement 1 · (⇒) proof 1.5 · (⇐) Cauchy sequence & limit A 2 · final 2/K < ε verification 1.5.

24 PYQ MAT-22 Q4 · also MAT-22 MCQ 17, MAT-21 MCQ 12, 16 · adapted

Dirichlet's function and h(x) = x + 1 on $\mathbb{Q} \cap [0,1], 0$ on irrationals, are not in $\mathcal{R}[0,1]$.

KNOW FIRST

T11 T10 P14

SKELETON

Set $\varepsilon_0 = 1/2 (or 1)$. For any $\eta > 0$, pick a partition P of [0,1] with $\|P\| < \eta$; tag P with rationals to get $\dot{P}$ and with irrationals to get $\dot{Q}$, and show $|S(\dot{P}) - S(\dot{Q})| \geq \varepsilon_0$.

PROOFS (BARTLE EXAMPLE 7.2.2(B) & EXERCISE 7.2.2)

Part 1: Dirichlet's function $f(x) = 1(x \in \mathbb{Q} \cap [0,1]), 0(x \in [0,1] \setminus \mathbb{Q})$ C7.

1. Put $\varepsilon_0 := \frac{1}{2} > 0$, and let $\eta > 0$ be arbitrary. Choose any partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of [0,1] with $\|P\| < \eta$.
   Set $\varepsilon_0 = 1/2$ to negate Theorem 7.2.1.

2. Since $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are both dense in $\mathbb{R}$, each subinterval $[x_{i-1}, x_i]$ contains a rational point t_i and an irrational point s_i. Let $\dot{P}$ be P tagged with the rational points t_i, and let $\dot{Q}$ be P tagged with the irrational points s_i.
   Density of rationals and irrationals (P14).

3. Then $S(f; \dot{P}) = \sum_{i=1}^n 1 \cdot (x_i - x_{i-1}) = 1$ and $S(f; \dot{Q}) = \sum_{i=1}^n 0 \cdot (x_i x_{i-1}) = 0$, so $|S(f; \dot{P}) - S(f; \dot{Q})| = |1 - 0| = 1 \geq \dfrac{1}{2} = \varepsilon_0$. Since $\eta > 0$ was arbitrary, the Cauchy Criterion 7.2.1 fails and $f \notin \mathcal{R}[0,1] (MAT-22$ MCQ 17 & MAT-21 MCQ 12 option B).
   Negation of Theorem 7.2.1.

Part 2 : $h(x) = x + 1(x \in \mathbb{Q} \cap [0,1]), 0(x \in [0,1] \setminus \mathbb{Q})$ (Bartle Ex 7.2.2; MAT-21 MCQ 16).

1. Put $\varepsilon_0 := 1 > 0$. For any $\eta > 0$ and the same $\dot{P}$ (rational tags $t_i \in [0,1]$) and $\dot{Q}$ (irrational tags s_i), we have $h(t_i) = t_i + 1 \geq 1$ and $h(s_i) = 0$, so $S(h;\dot{P}) = \sum_{i=1}^{n}(t_i+1)(x_i - x_{i-1}) \geq \sum_{i=1}^{n} 1 \cdot (x_i - x_{i-1}) = 1, \qquad S(h;\dot{Q}) = 0.$ Hence $|S(h;\dot{P}) - S(h;\dot{Q})| \geq 1 = \varepsilon_0$, so $h \notin \mathcal{R}[0,1]$ ($MAT-21$ MCQ 16 option B). ∎

$t_i \geq 0 \Rightarrow t_i + 1 \geq 1$ on [0,1].

Marks: Cauchy negation setup 1 · Dirichlet proof 2.5 · h(x)=x+1 proof 2.5.

25 PYQ MTS-26 Q18 also MTS-22 Q6, MAT-21 Q7

State and prove the Squeeze Theorem (7.2.3); apply it to h(x) = x on [0,1] (Example 7.2.6(b)).

KNOW FIRST

T12 T10 T14 Step functions are integrable P03

THEOREM 7.2.3 · SQUEEZE THEOREM (BARTLE P. 209)

Let f : $[a,b] \to \mathbb{R}$. Then $f \in \mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ there exist functions α_ε and ω_ε in $\mathcal{R}[a,b]$ with $\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x) \quad$ for all $x \in [a,b] \qquad$ and $\qquad \int_a^b(\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon.$

PROOF OF THEOREM 7.2.3 & EXAMPLE 7.2.6(B) (BARTLE PP. 209–211)

Proof of $7.2.3$: $(\Rightarrow)$ If $f \in \mathcal{R}[a,b]$, take $\alpha_\varepsilon = \omega_\varepsilon = f$; then $\int_a^b (\omega_\varepsilon \alpha_\varepsilon) = 0 < \varepsilon$. $(\Leftarrow)$

1. Let $\varepsilon > 0$. Since $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$, there exists $\delta_\varepsilon > 0$ such that any tagged partition $\dot{P}$ with $\|P\| < \delta_\varepsilon$ satisfies $|S(\alpha_\varepsilon; \dot{P}) - \int_a^b \alpha_\varepsilon| < \varepsilon$ and $|S(\omega_\varepsilon; \dot{P}) - \int_a^b \omega_\varepsilon| < \varepsilon$.
   Definition 7.1.1 applied to α_ε and ω_ε, taking $\min$ of the two deltas (M08).

2. Since $\alpha_\varepsilon \le f \le \omega_\varepsilon$, we have $S(\alpha_\varepsilon; \dot{P}) \le S(f; \dot{P}) \le S(\omega_\varepsilon; \dot{P})$, and therefore $\int_a^b \alpha_\varepsilon - \varepsilon < S(f; \dot{P}) < \int_a^b \omega_\varepsilon + \varepsilon$. If $\dot{Q}$ is any other tagged partition with $\|Q\| < \delta_\varepsilon$, then $\int_a^b \alpha_\varepsilon - \varepsilon < S(f; \dot{Q}) < \int_a^b \omega_\varepsilon + \varepsilon$ as well.
   Order preservation of Riemann sums (P02).

3. Subtracting these two inequalities and using Theorem 7.1.5 gives $|S(f; \dot{P}) - S(f; \dot{Q})| < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon = \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < \varepsilon + 2\varepsilon = 3\varepsilon$. Since $\varepsilon > 0$ is arbitrary, the Cauchy Criterion 7.2.1 implies $f \in \mathcal{R}[a,b]$.
   Theorem 7.1.5 and Cauchy Criterion 7.2.1 (T10).

Application to h(x) = x on [0,1] (Example 7.2.6(b)):

1 On the n-part equal partition $P_n = (0, \tfrac{1}{n}, \tfrac{2}{n}, \dots, 1)$, define step functions $\alpha_n(x) := \frac{k-1}{n}$ and $\omega_n(x) := \frac{k}{n}$ for $x \in [\frac{k-1}{n}, \frac{k}{n})(k = 1, \dots, n-1$, and closed on $[\frac{n-1}{n}, 1])$. Then $\alpha_n \le h \le \omega_n$ and by Theorem 7.2.5, $\int_0^1 \alpha_n = \frac{1}{n}\sum_{k=1}^{n} \frac{k-1}{n} = \frac{(n-1)n}{2n^2} = \frac{1}{2}(1\frac{1}{n}), \quad \int_0^1 \omega_n = \frac{1}{n}\sum_{k=1}^{n} \frac{k}{n} = \frac{n(n+1)}{2n^2} = \frac{1}{2}(1 + \frac{1}{n})$. Thus $\int_0^1 (\omega_n - \alpha_n) = \frac{1}{n} < \varepsilon$ for $n > 1/\varepsilon$, so $h \in \mathcal{R}[0,1]$ by 7.2.3, and $\frac{1}{2}(1 - \frac{1}{n}) \le \int_0^1 x\,dx \le \frac{1}{2}(1 + \frac{1}{n})\ \forall n \implies \int_0^1 x\,dx = \frac{1}{2}$. ∎

Sum of first n integers (P03) and Theorem 7.2.3.

ANSWER $\displaystyle\int_0^1 x\,dx = \frac{1}{2}.$

Marks: statement of 7.2.3 1 · proof of 7.2.3 3 · step functions αn, ωn and ∫01 x\,dx = 1/2 2.

26 PYQ MTS-26 Q5 also MAT-22 Q16

Every step function is integrable (Theorem 7.2.5); evaluate $\int_0^3 g$.

KNOW FIRST

T14 T13 T07 M06

THEOREM 7.2.5 · INTEGRABILITY OF STEP FUNCTIONS (BARTLE P. 210)

If $\varphi : [a,b] \to \mathbb{R}$ is a step function, then $\varphi \in \mathcal{R}[a,b]$. If $\varphi = \sum_{j=1}^{m} k_j \varphi_{J_j}$ where J_j has endpoints $c_j < d_j$, then $\int_a^b \varphi = \sum_{j=1}^{m} k_j (d_j - c_j)$.

PROOF OF THEOREM 7.2.5 (BARTLE PP. 210–211)

1. First, by Lemma 7.2.4, for any subinterval $J \subseteq [a,b]$ with endpoints c < d, the elementary step function $\varphi_J(x) = 1$ on J and 0 elsewhere belongs to $\mathcal{R}[a,b]$ with $\int_a^b \varphi_J = d - c (for J = [c,d], \|P\| < \delta \Rightarrow (d-c) - 2\delta \leq S(\varphi_{[c,d]}; \dot{P}) \leq (d-c) + 2\delta$, so $\delta_\varepsilon = \varepsilon/4$ works; the other three interval types differ at ≤ 2 endpoints, so Theorem 7.1.3 applies).
   Lemma 7.2.4 (T13).

2. By Definition 5.4.9, if $\varphi : [a,b] \to \mathbb{R}$ is an arbitrary step function, [a,b] is the union of finitely many non-overlapping subintervals $J_1, \ldots, J_m$ with endpoints $c_j \leq d_j$ such that φ has constant value k_j on J_j. Thus φ can be written as the linear combination $\varphi = \sum_{j=1}^m k_j \varphi_{J_j}$.
   Exercise 7.2.5 / M06.

3. By Lemma 7.2.4 and linearity of the integral (Theorem 7.1.5(a),(b), applied inductively to m terms), $\varphi \in \mathcal{R}[a,b]$ and $\int_a^b \varphi = \sum_{j=1}^m k_j \int_a^b \varphi_{J_j} = \sum_{j=1}^m k_j (d_j - c_j)$. ■
   Theorem 7.1.5(a),(b) (T07).

Example 7.2.6(a): For g(x) = 2 on [0,1] and g(x) = 3 on (1,3], we have $g = 2\varphi_{[0,1]} + 3\varphi_{(1,3]}$, so $\int_0^3 g = 2(1-0) + 3(3-1) = 2 + 6 = 8$.

ANSWER $\int_a^b \varphi = \sum_{j=1}^m k_j (d_j - c_j)$ **and** $\int_0^3 g = 8$.

Marks: Lemma 7.2.4 proof/statement 2 · decomposition ϕ = Σ kjϕ(Jj) 1.5 · linearity & formula 1.5 · ∫03 g = 8 1.

27 PYQ MAT-22 Q37b

Restriction to $[c,d] \subseteq [a,b]$ (Corollary 7.2.10) and subdivision formula (Corollary 7.2.11).

KNOW FIRST

T18 T17

COROLLARIES 7.2.10 & 7.2.11 (BARTLE P. 214)

If $f \in \mathcal{R}[a,b]$ and $[c,d] \subseteq [a,b]$, then $f|_{[c,d]} \in \mathcal{R}[c,d]$.
If $a = c_0 < c_1 < \cdots < c_m = b$, then $\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f$.

PROOFS

1. **Proof of Corollary 7.2.10:** Let $f \in \mathcal{R}[a,b]$ and $a \le c < d \le b$. If c = a, then $f|_{[c,b]} = f \in \mathcal{R}[c,b]$ by hypothesis; if c > a, then $c \in (a,b)$, so the Additivity Theorem $7.2.9(\Rightarrow)$ implies that the restriction $f|_{[c,b]}$ belongs to $\mathcal{R}[c,b]$.
   Additivity Theorem $7.2.9(\Rightarrow)$ at the cut point $c \in (a,b)$.

2. Next, consider $f|_{[c,b]} \in \mathcal{R}[c,b]$. If d = b, then $f|_{[c,d]} = f|_{[c,b]} \in \mathcal{R}[c,d]$; if d < b, then $d \in (c,b)$, so applying Theorem $7.2.9(\Rightarrow)$ to $f|_{[c,b]}$ on [c,b] at the interior point d shows that its restriction $f|_{[c,d]}$ belongs to $\mathcal{R}[c,d]$.
   Second application of Theorem $7.2.9(\Rightarrow)$ at $d \in (c,b)$.

3. **Proof of Corollary 7.2.11 (by induction on m):** By Corollary 7.2.10, each restriction $f|_{[c_{i-1},c_i]}$ belongs to $\mathcal{R}[c_{i-1},c_i]$. For $m = 1$, $\int_a^b f = \int_{c_0}^{c_1} f$ is immediate; for $m = 2$, $\int_a^b f = \int_{c_0}^{c_1} f + \int_{c_1}^{c_2} f$ is Theorem 7.2.9. Assume the formula holds for m - 1 subintervals. Given $a = c_0 < c_1 < \cdots < c_{m-1} < c_m = b$, apply Theorem 7.2.9 at $c = c_{m-1} \in (a,b)$ to get $\int_a^b f = \int_a^{c_{m-1}} f + \int_{c_{m-1}}^{c_m} f = \sum_{i=1}^{m-1} \int_{c_{i-1}}^{c_i} f + \int_{c_{m-1}}^{c_m} f = \sum_{i=1}^{m} \int_{c_{i-1}}^{c_i} f$. ■
   Induction hypothesis applied on $[a, c_{m-1}]$.

Marks: split at c 1.5 · split at d 1.5 · induction base & step for 7.2.11 3.

28 PYQ MAT-22 MCQ 24, 25 · adapted | also MAT-21 MCQ 18 · adapted

Bounded f integrable on every $[c,b](c \in (a,b))$ is in $\mathcal{R}[a,b]$ (Bartle Ex 7.2.11–7.2.13); application to $\sin(1/x)$.

KNOW FIRST

T18 T12 T17 T08

PROOF OF BARTLE EXERCISES 7.2.11–7.2.13

1. Assume $|f(x)| \leq M$ on [a,b] (M > 0) and $f|_{[c,b]} \in \mathcal{R}[c,b]$ for every $c \in (a,b)$. Given $\varepsilon > 0$, choose $c \in (a,b)$ such that c - a < $\frac{\varepsilon}{4M}$.
   Make the trouble strip [a,c] near a narrower than $\varepsilon/(4M)$.

2. Define $\alpha_c, \omega_c : [a,b] \to \mathbb{R}$ by $\alpha_c(x) :=$ -M and $\omega_c(x) := M$ for $x \in [a,c)$, and $\alpha_c(x) := \omega_c(x) := f(x)$ for $x \in [c,b]$. Since α_c, ω_c are constant on [a,c) and equal to $f \in \mathcal{R}[c,b]$ on [c,b], Theorem $7.2.9(\Leftarrow)$ and Theorem 7.1.3 imply $\alpha_c, \omega_c \in \mathcal{R}[a,b]$ and $\alpha_c(x) \leq f(x) \leq \omega_c(x)$ on [a,b].
   Additivity Theorem $7.2.9(\Leftarrow)$ glues the constant piece on [a,c] to f on [c,b].

3. Since $\omega_c - \alpha_c =$ 2M on [a,c) and 0 on [c,b], we have $\int_a^b(\omega_c - \alpha_c) = \int_a^c 2M + \int_c^b 0 = 2M(c-a)$ < $2M \cdot \frac{\varepsilon}{4M} = \frac{\varepsilon}{2} < \varepsilon$. By the Squeeze Theorem 7.2.3, $f \in \mathcal{R}[a,b]$.
   Squeeze Theorem 7.2.3 (T12).

4. Furthermore, for any $c \in (a,b)$, Theorem 7.2.9 gives $\int_a^b f - \int_c^b f = \int_a^c f$, so $|\int_a^b f - \int_c^b f| = |\int_a^c f| \leq M(c-a) \to 0$ as $c \to a^+$, proving $\lim_{c\to a^+} \int_c^b f = \int_a^b f$. (Thus in MAT-22 MCQ 24 / MAT-21 MCQ 18, the statement that does **not** hold is $(B) f \notin \mathcal{R}[a,b]$.)
   Exercise 7.1.8 bound $|\int_a^c f| \leq M(c-a)$ and Squeeze Theorem for limits.

5. **Application to $g(x) = \sin(1/x) (x \in (0,1]), g(0) = 0 (Ex 7.2.12$, MAT-22 MCQ 25):** We have $|g(x)| \leq 1$ on [0,1], and for each $c \in (0,1), g$ is continuous on [c,1] and hence in $\mathcal{R}[c,1]$ by Theorem 7.2.7. By Step $3, g \in \mathcal{R}[0,1] (MCQ 25$ option A).
   Apply Exercise 7.2.11 with M = 1.

6. **Why boundedness cannot be dropped (Ex 7.2.13):** f(x) = 1/x on (0,1], f(0) = 0 C16 is continuous on $[c,1] (so f \in \mathcal{R}[c,1])$ for every $c \in (0,1)$, but f is unbounded on [0,1], so $f \notin \mathcal{R}[0,1]$ by Theorem 7.1.6. ∎
   Counterexample C16.

Marks: αc, ωc construction & Squeeze 2.5 · limit ∫cb f → ∫ab f 1.5 · sin(1/x) ∈ $\mathscr{R}$[0,1] 1 · 1/x counterexample 1.

Section B · Paragraph / problem answers (Part 2: Q29–Q36)
6 marks each. Ceiling 72.

29 PYQ MTS-26 Q8

State and prove the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1).

KNOW FIRST

T20 T03 T18 P12 Mean Value Theorem (Bartle 6.2.4) M05

THEOREM 7.3.1 · FUNDAMENTAL THEOREM OF CALCULUS, FIRST FORM (BARTLE P. 216)

Suppose there is a finite set E in [a,b] and functions f, F : $[a,b] \to \mathbb{R}$ such that:
(a) F is continuous on [a,b];

(b) F'(x) = f(x) for all $x \in [a,b] \setminus E$;

(c) f belongs to $\mathcal{R}[a,b]$.

Then we have $\int_a^b f = F(b) - F(a)$.

SKELETON

For $E = \{a,b\}$, given $\varepsilon > 0$ choose δ_ε from $f \in \mathcal{R}[a,b]$ and a partition P with $\|P\| < \delta_\varepsilon$. Apply the MVT 6.2.4 to F on each $[x_{i-1}, x_i]$ to get $F(x_i)F(x_{i-1}) = f(u_i)(x_i - x_{i-1})$; sum to get $F(b) - F(a) = S(f; \dot{P}_u)$, which is within ε of $\int_a^b f$.

PROOF OF THEOREM 7.3.1 (BARTLE PP. 216–217)

1. We first prove the case where $E = \{a, b\}$. Let $\varepsilon > 0$ be given. Since $f \in \mathcal{R}[a, b]$ by hypothesis (c), there exists $\delta_\varepsilon > 0$ such that if $\dot{P}$ is any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$, then $|S(f; \dot{P}) - \int_a^b f| < \varepsilon$.
   Definition 7.1.1 (T03).

2. Let $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ be a partition of [a,b] with $\|P\| < \delta_\varepsilon$. Since F is continuous on $[x_{i-1}, x_i]$ by (a) and differentiable on $(x_{i-1}, x_i) \subseteq [a, b] \setminus E$ by (b), the Mean Value Theorem 6.2.4 (P12) applied to F on $[x_{i-1}, x_i]$ implies there exists $u_i \in (x_{i-1}, x_i)$ such that $F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}) = f(u_i)(x_i - x_{i-1}) \quad \text{for } i = 1, \ldots, n.$
   Mean Value Theorem 6.2.4 on $[x_{i-1}, x_i]$, using $u_i \notin E \Rightarrow F'(u_i) = f(u_i)$.

3. Adding these n equations and telescoping the left-hand side gives $F(b) - F(a) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) = \sum_{i=1}^n f(u_i)(x_i - x_{i-1}) = S(f; \dot{P}_u)$, where $\dot{P}_u := \{([x_{i-1}, x_i], u_i)\}_{i=1}^n$ is a tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$.
   Telescoping sum (M01, M05).

4. Substituting $S(f; \dot{P}_u) = F(b) - F(a)$ into Step 1 gives $|F(b) - F(a) \int_a^b f| < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\int_a^b f = F(b) - F(a)$.
   Arbitrary ε rule (P05).

5 **General finite set E (Exercise 7.3.1):** Let $a = c_0 < c_1 < \cdots < c_m = b$ be the points of $E \cup \{a, b\}$ in increasing order. On each $[c_{k-1}, c_k]$, F is continuous, F'(x) = f(x) on (c_{k-1}, c_k), and $f|_{[c_{k-1},c_k]} \in \mathcal{R}[c_{k-1}, c_k]$ by Corollary 7.2.10. Applying the first part on each subinterval and summing via Corollary 7.2.11 gives $\int_a^b f = \sum_{k=1}^m (F(c_k)F(c_{k-1})) = F(b) - F(a)$. ∎ Corollaries 7.2.10 and 7.2.11 (T18).

Marks: statement with (a),(b),(c) 1.5 · MVT step on [x(i-1),xi] 2 · telescoping to S(f;dot Pu) and conclusion 1.5 · general finite E 1.

30 PYQ MTS-25 Q6 also MAT-22 Q18, MAT-21 Q19

Substitution Theorem 7.3.8: statement, proof, five substitution evaluations, and $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$ via FTC 1.

KNOW FIRST

T25 T23 T20 T18 P11

THEOREM 7.3.8 · SUBSTITUTION THEOREM (BARTLE P. 220)

Let $J := [\alpha, \beta]$ and let $\varphi : J \to \mathbb{R}$ have a continuous derivative on J. If f : $I \to \mathbb{R}$ is continuous on an interval I containing $\varphi(J)$, then $\int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$.

PROOF OF THEOREM 7.3.8 (BARTLE EXERCISE 7.3.17, P. 224)

1. Define $F(u) := \int_{\varphi(\alpha)}^{u} f(x)\,dx$ for $u \in I$. Since f is continuous on I, Theorem 7.3.6 (FTC 2) implies that F'(u) = f(u) for all $u \in I$.
   Theorem 7.3.6 (T23).

2. Define $H(t) := F(\varphi(t))$ for $t \in J = [\alpha, \beta]$. By the Chain Rule 6.1.6, H is differentiable on J and $H'(t) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t)$ for all $t \in J$.
   Chain Rule 6.1.6 (P11).

3. Since f, φ, and φ' are continuous, $(f \circ \varphi)\varphi'$ is continuous on $[\alpha, \beta]$ and hence in $\mathcal{R}[\alpha, \beta]$ (Theorem 7.2.7). Since $H(\alpha) = F(\varphi(\alpha)) = 0$, Theorem 7.3.1 (FTC 1 with $E = \varnothing$) gives $\int_{\alpha}^{\beta} f(\varphi(t))\,\varphi'(t)\,dt = H(\beta) - H(\alpha) = F(\varphi(\beta)) = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$. ■
   Theorem 7.2.7 and Theorem 7.3.1 (T20).

Five substitution evaluations (Example 7.3.9(a) & Exercise 7.3.18):

- 1. $\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt (MTS-25$ **Q6, MAT-21 Q19):** Put $\varphi(t) = \sqrt{t}$ on $[1,4] (\varphi'(t) = \frac{1}{2\sqrt{t}}$ continuous on [1,4]) and $f(x) = 2\sin x$: $\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt = \int_1^2 2\sin x\,dx = [-2\cos x]_1^2 = 2(\cos 1 - \cos 2) \approx 1.912898.$
- 2. $\int_1^4 \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt (Ex 7.3.18(d))$**:** Put $\varphi(t) = \sqrt{t}$ on [1,4] and $f(x) = 2\cos x$: $\int_1^4 \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt = \int_1^2 2\cos x\,dx = [2\sin x]_1^2 = 2(\sin 2 - \sin 1) \approx 0.135653.$

- 3. $\int_0^1 t\sqrt{1+t^2}\,dt (Ex 7.3.18(a))$: Put $\varphi(t) = 1+t^2$ on $[0,1] (\varphi'(t) = 2t$ continuous) and $f(x) = \frac{1}{2}\sqrt{x}$: $\int_0^1 t\sqrt{1+t^2}\,dt = \int_1^2 \frac{1}{2}x^{1/2}\,dx = [\frac{1}{3}x^{3/2}]_1^2 = \frac{2\sqrt{2}-1}{3} \approx 0.609476.$
- 4. $\int_0^2 t^2(1+t^3)^{-1/2}\,dt (Ex 7.3.18(b))$: Put $\varphi(t) = 1+t^3$ on $[0,2] (\varphi'(t) = 3t^2$ continuous) and $f(x) = \frac{1}{3}x^{-1/2}$: $\int_0^2 t^2(1+t^3)^{-1/2}\,dt = \int_1^9 \frac{1}{3}x^{-1/2}\,dx = [\frac{2}{3}\sqrt{x}]_1^9 = \frac{2}{3}(3-1) = \frac{4}{3}.$
- 5. $\int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}}\,dt (Ex 7.3.18(c))$: Put $\varphi(t) = 1+\sqrt{t}$ on $[1,4] (\varphi'(t) = \frac{1}{2\sqrt{t}}$ continuous on [1,4]) and $f(x) = 2\sqrt{x}$: $\int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}}\,dt = \int_2^3 2x^{1/2}\,dx = [\frac{4}{3}x^{3/2}]_2^3 = \frac{4}{3}(3\sqrt{3}-2\sqrt{2}) \approx 3.156967.$

Why 7.3.8 fails on [0,4] and how $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$ is evaluated (Example 7.3.9(b); PYQ MAT-22 Q18): On $[0,4], \varphi(t) = \sqrt{t}$ has $\varphi'(t) = \frac{1}{2\sqrt{t}}$, which is undefined and unbounded at t = 0, so Theorem 7.3.8 does not apply. Instead, let $g(t) := \frac{\sin\sqrt{t}}{\sqrt{t}}$ on (0,4] and g(0) := 1. Since $|\sin u| \le |u|, |g(t)| \le 1$ is bounded on [0,4] and continuous on [c,4] for every $c \in (0,4)$, so $g \in \mathcal{R}[0,4]$ by Exercise 7.2.11 (Q28). Define $F(t) := -2\cos\sqrt{t}$ on [0,4]; then F is continuous on [0,4] and F'(t) = g(t) for all $t \in (0,4] (E = \{0\})$. By FTC 1 (Theorem 7.3.1), $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt = F(4) - F(0) = -2\cos 2 - (-2\cos 0) = 2(1-\cos 2) \approx 2.832294.$

ANSWER $\int_1^4 \dfrac{\sin\sqrt{t}}{\sqrt{t}}\,dt = 2(\cos 1 - \cos 2) \approx 1.912898$; $\int_0^4 \dfrac{\sin\sqrt{t}}{\sqrt{t}}\,dt = 2(1 - \cos 2) \approx 2.832294$.

Marks: statement & proof of 7.3.8 2 · substitution evaluations 2 · [0,4] explanation & FTC 1 evaluation $2(1-\cos 2)$2.

31 PYQ MAT-22 Q6

State Lebesgue's Integrability Criterion (7.3.12) and apply it to five benchmark functions (Examples 7.3.13(a)–(e)).

KNOW FIRST

T27 T26 T08

THEOREM 7.3.12 · LEBESGUE'S INTEGRABILITY CRITERION (BARTLE P. 221)

A bounded function f : $[a, b] \to \mathbb{R}$ is Riemann integrable if and only if it is continuous almost everywhere on [a,b] (equivalently, the set D(f) of points of discontinuity of f on [a,b] is a null set).

SKELETON

For each function on [0,1], check (1) whether it is bounded, and (2) whether its discontinuity set D is a null set.

Applications (Bartle Examples 7.3.13(a)–(e)):

- **(a) Any step function φ on [0,1]:** φ takes finitely many values, so it is bounded, and it is discontinuous at most at the finitely many endpoints of its constant intervals. Since every finite set is a null set, $\varphi \in \mathcal{R}[0, 1]$.
- **(b) Any monotone function f on [0,1]:** f is bounded between f(0) and f(1), and by Theorem 5.6.4 (P10) its set of discontinuities is at most countable, hence a null set (T26). By Theorem $7.3.12, f \in \mathcal{R}[0, 1]$.
- $(c) G(1/n) = 1/n (n \in \mathbb{N}), G(x) = 0$ **elsewhere on [0,1] (Example 7.1.4(d)):** G is bounded $(0 \leq G(x) \leq 1)$, continuous at x = 0 (since

$\lim_{x\to 0} G(x) = 0 = G(0)$) and at every point outside $\{1/n : n \in \mathbb{N}\}$, and discontinuous precisely on the countable set $D = \{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\}$. Since every countable set is a null set, $G \in \mathcal{R}[0,1]$.

- **(d) Dirichlet's function f on [0,1]** C7 **:** f is bounded $(0 \le f \le 1)$, but f is discontinuous at **every** point of [0,1]. Since D = [0,1] is not a null set, $f \notin \mathcal{R}[0,1]$.
- **(e) Thomae's function h on [0,1]** C8 **:** h is bounded $(0 \le h \le 1)$, continuous at every irrational in [0,1], and discontinuous precisely on $D = \mathbb{Q} \cap [0,1]$. By Example $7.3.11$, $\mathbb{Q} \cap [0,1]$ is a null set, so $h \in \mathcal{R}[0,1]$.

Marks: statement of Theorem 7.3.12 1 · 1 mark for each of the five cases (a)–(e).

32 PYQ MAT-21 Q4

State and prove the Composition Theorem (7.3.14); counterexample when φ is not continuous.

KNOW FIRST

T28 T27 T26 P07 Continuous on [a,b]: Boundedness, Max–Min, IVT

THEOREM 7.3.14 · COMPOSITION THEOREM (BARTLE P. 222)

Let $f \in \mathcal{R}[a,b]$ with $f([a,b]) \subseteq [c,d]$, and let $\varphi : [c,d] \to \mathbb{R}$ be continuous. Then the composition $\varphi \circ f$ belongs to $\mathcal{R}[a,b]$.

PROOF OF THEOREM 7.3.14 (BARTLE P. 222)

1. If f is continuous at a point $u \in [a,b]$, then since φ is continuous at $f(u) \in [c,d]$, the composite function $\varphi \circ f$ is also continuous at u.
   Composition of continuous functions is continuous (P07).

2. Consequently, the set D of points of discontinuity of $\varphi \circ f$ on [a,b] is contained in the set D(f) of points of discontinuity of f.
   Contrapositive of Step 1 : $u \in D \Rightarrow u \in D(f)$.

3. Since $f \in \mathcal{R}[a,b]$, Lebesgue's Integrability Criterion 7.3.12 implies that D(f) is a null set. Since any subset of a null set is a null set (T26), D is a null set.
   Theorem $7.3.12(\Rightarrow)$ and subset property of null sets.

4. Moreover, since φ is continuous on the closed bounded interval $[c,d]$, φ is bounded on [c,d] (Theorem 5.3.2), so $\varphi \circ f$ is bounded on [a,b]. By Lebesgue's Integrability Criterion $7.3.12$, $\varphi \circ f \in \mathcal{R}[a,b]$. ∎
   Boundedness Theorem 5.3.2 (P07) and Theorem $7.3.12(\Leftarrow)$.

Counterexample showing continuity of φ cannot be dropped (Exercise 7.3.22): Let $h \in \mathcal{R}[0,1]$ be Thomae's function C8, and define $\varphi : [0,1] \to \mathbb{R}$ by $\varphi(0) := 0$ and $\varphi(y) := 1$ for $y \in (0,1]$. Note that $\varphi \in \mathcal{R}[0,1]$ (step function, discontinuous only at 0). However, for $x \in [0,1]$, $(\varphi \circ h)(x) = 1$ if $x \in \mathbb{Q} \cap [0,1]$ (where h(x) = 1/n > 0) and $(\varphi \circ h)(x) = \varphi(0) = 0$ if $x \in$

$[0,1] \setminus \mathbb{Q}$: thus $\varphi \circ h$ is Dirichlet's function C7, which is **not** in $\mathcal{R}[0,1]$.

Marks: statement 1· $D(\varphi \circ f) \subseteq D(f)$ null 2 · boundedness & Lebesgue conclusion 1.5 · counterexample 1.5.

33 PYQ MTS-25 Q8 also MTS-22 Q7, MAT-21 Q20

If $f \in \mathcal{R}[a,b]$, then $|f| \in \mathcal{R}[a,b]$ and $|\int_a^b f| \le \int_a^b |f| \le M(b-a)$ (Corollary 7.3.15); converse false.

KNOW FIRST

T29 T28 T08 T07

COROLLARY 7.3.15 (BARTLE P. 222)

Suppose $f \in \mathcal{R}[a,b]$. Then its absolute value $|f|$ is in $\mathcal{R}[a,b]$, and $|\int_a^b f| \le \int_a^b |f| \le M(b-a)$, where $|f(x)| \le M$ for all $x \in [a,b]$.

PROOF OF COROLLARY 7.3.15 (BARTLE P. 222)

1. Since $f \in \mathcal{R}[a,b]$, Theorem 7.1.6 implies that f is bounded on [a,b], say $|f(x)| \le M$ for all $x \in [a,b]$, so $f([a,b]) \subseteq [-M, M]$.
   Boundedness Theorem 7.1.6 (T08).

2. The function $\varphi(t) := |t|$ is continuous on [-M,M]. By the Composition Theorem 7.3.14, $|f| = \varphi \circ f$ belongs to $\mathcal{R}[a,b]$.
   Composition Theorem 7.3.14 (T28).

3. Since $-|f(x)| \le f(x) \le |f(x)|$ for all $x \in [a,b]$, Theorem 7.1.5(a) and (c) give -\int_a^b \lvert f\rvert \le \int_a^b f \le \int_a^b \lvert f\rvert, \quad \text{and\ hence} $|\int_a^b f| \le \int_a^b |f|$.
   Linearity and monotonicity of the integral (7.1.5a,c, T07) and P05.

4. Since $|f(x)| \le M$ for all $x \in [a,b]$, Theorem 7.1.5(c) and Example 7.1.4(a) give $\int_a^b |f| \le \int_a^b M = M(b-a)$. ∎
   Constant integral (T05).

Converse is false: Define f : $[0,1] \to \mathbb{R}$ by f(x) := 1 if $x \in \mathbb{Q} \cap [0,1]$ and f(x) := -1 if $x \in [0,1] \setminus \mathbb{Q}$. Then $|f(x)| = 1$ for all $x \in [0,1]$, so $|f| \in \mathcal{R}[0,1]$, but $f \notin \mathcal{R}[0,1]$ (rational tags give S = 1, irrational tags give S = -1).

Marks: | f| ∈ ℛ[a,b] via 7.1.6 & 7.3.14 2 · |∫ f| ≤ ∫| f| ≤ M(b-a) 2.5 · counterexample for converse 1.5.

34 PYQ MTS-26 Q6 also MAT-22 Q17, MAT-21 Q18

State and prove the Product Theorem (7.3.16); show $\int_a^b fg \neq (\int_a^b f)(\int_a^b g)$ in general.

KNOW FIRST

T30 T28 T08 T07

THEOREM 7.3.16 · PRODUCT THEOREM (BARTLE P. 222)

If f and g belong to $\mathcal{R}[a,b]$, then the product fg belongs to $\mathcal{R}[a,b]$.

PROOF OF THEOREM 7.3.16 (BARTLE P. 222)

1. Since $f, g \in \mathcal{R}[a,b]$, Theorem 7.1.5(b) implies $f+g \in \mathcal{R}[a,b]$. By the Boundedness Theorem 7.1.6, f, g, and f+g are bounded on [a,b]; choose M > 0 so that f([a,b]), g([a,b]), and (f+g)([a,b]) are all contained in [-M,M].
   Theorem 7.1.5(b) and Boundedness Theorem 7.1.6 (T08).

2. The squaring function $\varphi(t) := t^2$ is continuous on [-M,M]. By the Composition Theorem 7.3.14, $f^2 = \varphi \circ f$, $g^2 = \varphi \circ g$, and $(f+g)^2 = \varphi \circ (f+g)$ all belong to $\mathcal{R}[a,b]$.
   Composition Theorem 7.3.14 (T28) with $\varphi(t) = t^2$.

3. Since we have the algebraic identity $fg = \frac{1}{2}[(f+g)^2 - f^2 - g^2]$, Theorem 7.1.5(a) and (b) imply that $fg \in \mathcal{R}[a,b]$. ∎
   Linearity of the integral (Theorem 7.1.5, T07).

Why $\int_a^b fg \neq (\int_a^b f)(\int_a^b g)$ in general: Let f(x) = g(x) = x on [0,1]. Then $\int_0^1 fg = \int_0^1 x^2\,dx = \frac{1}{3}$, whereas $(\int_0^1 f)(\int_0^1 g) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \neq \frac{1}{3}$.

Marks: statement $1\cdot$ $f^2, g^2, (f+g)^2 \in \mathcal{R}[a,b]$ via 7.3.14 2.5 $\cdot$ polarisation identity & 7.1.5 1.5 $\cdot$ counterexample for integral of product 1.

35

State and prove Integration by Parts (Theorem 7.3.17); evaluate $\int_0^1 xe^x\,dx$.

KNOW FIRST

T31 T30 T20 Fundamental Theorem of Calculus (First Form) P11

THEOREM 7.3.17 · INTEGRATION BY PARTS (BARTLE P. 222, EQ. (7))

Let F, G be differentiable on [a,b] and let f := F' and g := G' belong to $\mathcal{R}[a,b]$. Then $\int_a^b$ fG = $[FG]_a^b - \int_a^b$ Fg.

PROOF OF THEOREM 7.3.17 (BARTLE P. 223)

1. By the Product Rule 6.1.3(c) (P11), the product FG is differentiable on [a,b] and $(FG)'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x) \quad \text{for all } x \in [a,b]$.
   Product Rule 6.1.3(c) (P11).

2. Since F and G are differentiable on [a,b], they are continuous on [a,b] (Theorem 6.1.2) and hence belong to $\mathcal{R}[a,b]$ (Theorem 7.2.7). Since $f, g \in \mathcal{R}[a,b]$ by hypothesis, the Product Theorem 7.3.16 implies that fG $\in \mathcal{R}[a,b]$ and Fg $\in \mathcal{R}[a,b]$, so (FG)' = fG + Fg $\in \mathcal{R}[a,b]$ by Theorem 7.1.5(b).
   Theorems 6.1.2, 7.2.7, 7.3.16 (T30), and 7.1.5(b).

3. Applying the First Form of the Fundamental Theorem 7.3.1 (with $E = \varnothing$) to FG gives $[FG]_a^b = \int_a^b (FG)' = \int_a^b \text{fG} + \int_a^b \text{Fg}$, $\quad \text{whence} \quad \int_a^b \text{fG} = [FG]_a^b - \int_a^b \text{Fg}$. ■
   Theorem 7.3.1 (T20) and Theorem 7.1.5(b).

Evaluation of $\int_0^1 xe^x\,dx$: Put $F(x) := e^x (f(x) = F'(x) = e^x \in \mathcal{R}[0,1])$ and $G(x) := x (g(x) = G'(x) = 1 \in \mathcal{R}[0,1])$. By Theorem 7.3.17, $\int_0^1 e^x \cdot x\,dx = [xe^x]_0^1 - \int_0^1 e^x \cdot 1\,dx = (1 \cdot e - 0)[e^x]_0^1 = e - (e-1) = 1$.

ANSWER $\displaystyle\int_0^1 xe^x\,dx = 1.$

Marks: statement 1 · Product Rule + Product Theorem step 2 · FTC 1 conclusion 1.5 · ∫01 x ex\,dx = 1 1.5.

36 PYQ MTS-25 Q19 also MAT-22 Q32

State and prove Taylor's Theorem with the Integral Remainder (7.3.18); verify for $f(x) = e^x$ on [0,1] with n = 1, 2.

KNOW FIRST

T32 Taylor's Theorem with the Integral Remainder T31

T20 Fundamental Theorem of Calculus (First Form)

THEOREM 7.3.18 · TAYLOR'S THEOREM WITH THE REMAINDER (BARTLE P. 223, EQS. (8)–(9))

Suppose that f', . . . , $f^{(n)}$, $f^{(n+1)}$ exist on [a,b] and that $f^{(n+1)} \in \mathcal{R}[a,b]$. Then we have $f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + R_n$, where the remainder is given by $R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\,(b-t)^n\,dt$.

PROOF OF THEOREM 7.3.18 (BARTLE P. 223, WITH MISPRINT CORRECTED)

1. Apply Integration by Parts (Theorem 7.3.17) to R_n with $F(t) := f^{(n)}(t) (so f(t) = F'(t) = f^{(n+1)}(t) \in \mathcal{R}[a,b])$ and $G(t) := (b-t)^n/n! (so g(t) = G'(t) = -(b-t)^{n-1}/(n-1)! \in \mathcal{R}[a,b])$.
   Check hypotheses of Theorem 7.3.17 (T31).

2. Since G(b) = 0 and $G(a) = (b-a)^n/n!$, Theorem 7.3.17 gives $R_n = [\frac{f^{(n)}(t)(b-t)^n}{n!}]_{t=a}^{t=b} - \int_a^b f^{(n)}(t)(-\frac{(b-t)^{n-1}}{(n-1)!})dt = -\frac{f^{(n)}(a)}{n!}(b-a)^n + R_{n-1}$, where $R_{n-1} = \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\,(b-t)^{n-1}\,dt$.
   Boundary evaluation at t=b and t=a. (Note: Bartle p. 223 prints $(b-a)^{n-1}$ in the first displayed integral; the correct integrand is $(b-t)^{n-1}$.)

3. For each $k \le n$, $f^{(k)}$ is differentiable on [a,b], hence continuous (Theorem 6.1.2) and in $\mathcal{R}[a,b]$ (Theorem 7.2.7). Repeating Step 2 n times yields $R_{k-1} = \frac{f^{(k)}(a)}{k!}(b-a)^k + R_k$ down to $R_0 = \int_a^b f'(t)\,dt = f(b) - f(a) (by$ FTC 1, Theorem 7.3.1), which rearranges to Taylor's formula. ∎
   Repeated Integration by Parts and FTC 1 (T20).

Verification for $f(x) = e^x, a = 0, b = 1$: Here $f^{(k)}(0) = 1$ for all k. For $n = 1 : T_1 = 1 + 1 = 2$ and $R_1 = \int_0^1 e^t(1-t)\,dt = [e^t(1-t)]_0^1 + \int_0^1 e^t\,dt$ = -1 +$(e-1) = e - 2 \approx 0.718282$, so $T_1 + R_1 = e$. For $n = 2 : T_2 = 1 +$

$1 + \frac{1}{2} = \frac{5}{2}$ and $R_2 = -\frac{1}{2} + R_1 = e - \frac{5}{2} \approx 0.218282$, so $T_2 + R_2 = e$.

ANSWER $R_1 = e - 2 \approx 0.718282; \quad R_2 = e - \dfrac{5}{2} \approx 0.218282.$

Marks: statement 1.5 · IBP recurrence Rn = -frac(f((n))(a))(n!)(b-a)n + R(n-1) 2 · induction to R0 = f(b)-f(a) 1 · ex check 1.5.

Section C · Essay answers (Q37–Q40)

10 marks each. Answer any two.

37 PYQ MTS-25 Q18 also MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, MAT-21 MCQ 15

Continuous on $[a,b] \Rightarrow$ Riemann integrable (Theorem 7.2.7); where hypotheses are used; converses.

KNOW FIRST

T15 T12 T14 P07 Continuous on [a,b]: Boundedness, Max–Min, IVT P08

THEOREM 7.2.7 · INTEGRABILITY OF CONTINUOUS FUNCTIONS (BARTLE P. 212)

If f : $[a,b] \to \mathbb{R}$ is continuous on [a,b], then $f \in \mathcal{R}[a,b]$.

SKELETON

1. Use UCT 5.4.3 to get $\delta_\varepsilon > 0$ so $|u-v| < \delta_\varepsilon \Rightarrow |f(u)-f(v)| < \varepsilon/(b-a)$.2. Pick a partition P with $\|P\| < \delta_\varepsilon$.3. Use Max–Min 5.3.4 on each I_i to get $u_i, v_i \in I_i$ with $f(u_i) = \min_{I_i} f, f(v_i) = \max_{I_i} f$.4. Define step functions $\alpha_\varepsilon, \omega_\varepsilon$ with heights $f(u_i), f(v_i)$ and apply the Squeeze Theorem 7.2.3.

PROOF OF THEOREM 7.2.7 (BARTLE P. 212)

1. Since f is continuous on the closed bounded interval [a,b], the Uniform Continuity Theorem 5.4.3 (P08) implies that f is uniformly continuous on [a,b]. Therefore, given any $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that if $u, v \in [a,b]$ and $|u - v| < \delta_\varepsilon$, then $|f(u) - f(v)| < \dfrac{\varepsilon}{b-a}$.
   Uniform Continuity Theorem 5.4.3 applied with tolerance $\varepsilon/(b-a) > 0$.

2. Let $P = \{I_i\}_{i=1}^n$ be a partition of [a,b] with $\|P\| < \delta_\varepsilon$, where $I_i = [x_{i-1}, x_i]$.
   Choose an equal partition with $n > (b-a)/\delta_\varepsilon$.

3. Since f is continuous on each closed bounded subinterval $I_i = [x_{i-1}, x_i]$, the Maximum–Minimum Theorem 5.3.4 (P07) implies that there exist points $u_i, v_i \in I_i$ where f attains its minimum and maximum values on I_i, respectively: $f(u_i) \le f(x) \le f(v_i) \quad \text{for all } x \in I_i$.
   Maximum–Minimum Theorem 5.3.4 on each I_i.

4. Define step functions $\alpha_\varepsilon, \omega_\varepsilon : [a,b] \to \mathbb{R}$ by $\alpha_\varepsilon(x) := f(u_i)$ for $x \in [x_{i-1}, x_i)$ $(i < n)$, $\alpha_\varepsilon(x) := f(u_n)$ for $x \in [x_{n-1}, x_n]$, $\omega_\varepsilon(x) := f(v_i)$ for $x \in [x_{i-1}, x_i)$ $(i < n)$, $\omega_\varepsilon(x) := f(v_n)$ for $x \in [x_{n-1}, x_n]$. By Theorem 7.2.5, $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$, and by Step 3, $\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x)$ for all $x \in [a,b]$.
   Step-function integrability (Theorem 7.2.5, T14).

5. Since $u_i, v_i \in I_i$ and $\|P\| < \delta_\varepsilon$, we have $|u_i v_i| \leq x_i - x_{i-1} < \delta_\varepsilon$, so $f(v_i) - f(u_i) < \dfrac{\varepsilon}{b-a}$ by Step 1. Therefore, by formula (5) of Theorem 7.2.5, $0 \leq \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n (f(v_i) - f(u_i))(x_i - x_{i-1}) < \sum_{i=1}^n \dfrac{\varepsilon}{b-a}(x_i - x_{i-1}) = \dfrac{\varepsilon}{b-a}(b-a) = \varepsilon$.
   Theorem 7.2.5 formula and telescoping $\sum(x_i - x_{i-1}) = b - a (M01)$.

6. Since $\varepsilon > 0$ is arbitrary, the Squeeze Theorem 7.2.3 implies that $f \in \mathcal{R}[a,b]$. ∎
   Squeeze Theorem 7.2.3 (T12).

Where each hypothesis was used: Continuity of f and the fact that [a,b] is closed and bounded are used twice: (i) in Step 1 to invoke the Uniform Continuity Theorem 5.4.3 so a single δ_ε works across all subintervals, and (ii) in Step 3 to invoke the Maximum–Minimum Theorem 5.3.4 on each closed subinterval I_i.

Converses are FALSE (MAT-22 MCQ 21, MAT-21 MCQ 15): Option (A) "f continuous on $[a,b] \Rightarrow f \in \mathcal{R}[a,b]$" is true, while (B) and (C) are false: the step function g(x) = 2 on [0,1], 3 on (1,3] (or Thomae's function h C8) belongs to $\mathcal{R}$, yet is discontinuous and hence not uniformly continuous.

Marks: UCT step 2 · Max–Min step 2 · step functions αε, ωε 2 · ∫(ωε-αε) < ε & Squeeze 2 · where hypotheses used & converses 2.

38 PYQ MTS-26 Q19 also MAT-22 Q30, MAT-22 MCQ 22

Monotone on $[a,b] \Rightarrow$ Riemann integrable (Theorem 7.2.8); staircase calculation for $f(x) = x^2$ on [0,1] (n=4); converse.

KNOW FIRST

T16 T12 T14 M05

THEOREM 7.2.8 · INTEGRABILITY OF MONOTONE FUNCTIONS (BARTLE P. 212)

If f : $[a,b] \to \mathbb{R}$ is monotone on [a,b], then $f \in \mathcal{R}[a,b]$.

SKELETON

Assume f is increasing on [a,b]. Partition [a,b] into n equal subintervals of length (b-a)/n. Define step functions $\alpha(x) = f(x_{k-1})$ and $\omega(x) = f(x_k)$ on I_k, telescope $\int_a^b (\omega - \alpha) = \dfrac{b-a}{n}(f(b) - f(a))$, choose $n > (b-a)(f(b) - f(a))/\varepsilon$, and apply the Squeeze Theorem 7.2.3.

PROOF OF THEOREM 7.2.8 (BARTLE P. 212)

1. Assume f is increasing on I = [a,b] (if f is decreasing, then -f is increasing, so -f $\in \mathcal{R}[a,b]$ and hence $f = -(-f) \in \mathcal{R}[a,b]$ by Theorem 7.1.5(a)). For $n \in \mathbb{N}$, partition [a,b] into n equal subintervals $I_k = [x_{k-1}, x_k]$ with $x_k - x_{k-1} = \dfrac{b-a}{n} (k = 1, \ldots, n)$.
   Equal subinterval lengths (b-a)/n allow the step integrals to telescope.

2. Since f is increasing on $I_k = [x_{k-1}, x_k]$, its minimum on I_k is $f(x_{k-1})$ and its maximum on I_k is $f(x_k)$: $f(x_{k-1}) \leq f(x) \leq f(x_k) \quad \text{for all } x \in I_k$.
   Definition of an increasing function (P10).

3. Define step functions $\alpha, \omega : [a,b] \to \mathbb{R}$ by $\alpha(x) := f(x_{k-1})$ and $\omega(x) := f(x_k)$ for $x \in [x_{k-1}, x_k) (k = 1, \ldots, n-1)$, and $\alpha(x) := f(x_{n-1}), \omega(x) := f(x_n)$ for $x \in [x_{n-1}, x_n]$. Then $\alpha, \omega \in \mathcal{R}[a,b]$ by Theorem 7.2.5 and $\alpha(x) \leq f(x) \leq \omega(x)$ for all $x \in [a,b]$.
   Theorem 7.2.5 (T14).

4. By Theorem 7.2.5, $\int_a^b \alpha = \dfrac{b-a}{n}(f(x_0) + f(x_1) + \cdots + f(x_{n-1})), \qquad \int_a^b \omega = \dfrac{b-a}{n}(f(x_1) + \cdots + f(x_{n-1}) + f(x_n))$.
   Subtracting and cancelling the interior terms $f(x_1), \ldots, f(x_{n-1})$ yields $\int_a^b (\omega - \alpha) = \dfrac{b-a}{n}(f(x_n) - f(x_0)) = \dfrac{b-a}{n}(f(b)f(a))$.
   Telescoping sum (M01, M05) with $x_0 = a$ and $x_n = b$.

5 Given $\varepsilon > 0$, choose $n \in \mathbb{N}$ with $n > \frac{(b-a)(f(b)f(a))}{\varepsilon}$. Then $\int_a^b (\omega - \alpha) < \varepsilon$, so by the Squeeze Theorem 7.2.3, $f \in \mathcal{R}[a,b]$. ∎

Archimedean property (P13) and Squeeze Theorem 7.2.3 (T12).

Staircase illustration for $f(x) = x^2$ on [0,1] with n = 4: Here $P_4 = (0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1)$ has width $\Delta x_k = \frac{1}{4}$, and $f(x) = x^2$ takes values $0, \frac{1}{16}, \frac{4}{16}, \frac{9}{16}, 1$ at the partition points:

$\int_0^1 \alpha = \frac{1}{4}(0 + \frac{1}{16} + \frac{4}{16} + \frac{9}{16}) = \frac{14}{64} = \frac{7}{32} = 0.21875, \quad \int_0^1 \omega = \frac{1}{4}(\frac{1}{16} + \frac{4}{16} + \frac{9}{16} + 1) = \frac{30}{64} = \frac{15}{32} = 0.46875,$

and $\int_0^1 (\omega - \alpha) = 0.46875 - 0.21875 = 0.25 = \frac{1-0}{4}(1-0)$.

Converse is FALSE (MAT-22 MCQ 22): Option (A) is true, while (B) "f $\in \mathcal{R}[a,b] \Rightarrow f$ is monotone" is false: $f(x) = \sin x$ on $[0, \pi] (or x^2$ on [-1,1], or Thomae's function C8) is in $\mathcal{R}$ but not monotone.

ANSWER **For x^2 on $[0,1] (n = 4)$: $\int_0^1 \alpha = 0.21875, \int_0^1 \omega = 0.46875$, difference = 0.25.**

Marks: equal partition & α,ω setup 2.5 · telescoping ∫(ω-α) = (b-a)/(n)(f(b)-f(a)) 2.5 · choice of n & Squeeze 2 · x2 (n=4) calculation 2 · converse false 1.

39 PYQ MTS-26 Q25 also MAT-22 Q37a, MTS-22 Q25

State and prove the Additivity Theorem (Theorem 7.2.9).

KNOW FIRST

T17 T10 T08 T03 M08 The $\varepsilon/2 (or \varepsilon/3)$ argument and $\delta = \min\{\delta_1, \delta_2\}$

THEOREM 7.2.9 · ADDITIVITY THEOREM (BARTLE P. 213)

Let f : $[a, b] \to \mathbb{R}$ and let $c \in (a, b)$. Then $f \in \mathcal{R}[a, b]$ if and only if its restrictions to [a,c] and to [c,b] are both Riemann integrable. In this case $\int_a^b f = \int_a^c f + \int_c^b f$.

SKELETON

($\Leftarrow$) Take δ', δ'' for f_1, f_2 with $\varepsilon/3$; state f is bounded by M > 0 (7.1.6 on each piece); put $\delta_\varepsilon = \min\{\delta', \delta'', \varepsilon/(6M)\}$. Split into Case $(i) c \in Q$ and Case $(ii) c \in (x_{k-1}, x_k)$.($\Rightarrow$) Use Cauchy 7.2.1 by extending $\dot{P}_1, \dot{Q}_1$ on [a,c] with the same $\dot{R}$ on [c,b].

PROOF OF THEOREM 7.2.9 (BARTLE PP. 213–214)

($\Leftarrow$)

1. Suppose $f_1 := f|_{[a,c]} \in \mathcal{R}[a,c]$ with integral L_1 and $f_2 := f|_{[c,b]} \in \mathcal{R}[c,b]$ with integral L_2. Given $\varepsilon > 0$, choose $\delta' > 0$ such that $\|P_1\| < \delta' \Rightarrow |S(f_1; \dot{P}_1) - L_1| < \varepsilon/3$, and choose $\delta'' > 0$ such that $\|P_2\| < \delta'' \Rightarrow |S(f_2; \dot{P}_2) - L_2| < \varepsilon/3$.
   Definition 7.1.1 applied on [a,c] and [c,b] with tolerance $\varepsilon/3$.

2. Since $f_1 \in \mathcal{R}[a,c]$ and $f_2 \in \mathcal{R}[c,b]$, Theorem 7.1.6 implies that f is bounded on [a,c] and on [c,b], and hence bounded on [a,b]. Choose M > 0 such that $|f(x)| \leq M$ for all $x \in [a,b] (if f \equiv 0$, take any M > 0), and define $\delta_\varepsilon := \min\{\delta',\, \delta'',\, \frac{\varepsilon}{6M}\} > 0$. Let $\dot{Q} = \{(I_i, t_i)\}_{i=1}^m$ be any tagged partition of [a,b] with $\|Q\| < \delta_\varepsilon$.
   Boundedness Theorem 7.1.6 (T08) and three-way minimum (M08).

3. **Case (i): c is a partition point of $\dot{Q}$.** Split $\dot{Q}$ at c into a tagged partition $\dot{Q}_1$ of [a,c] and a tagged partition $\dot{Q}_2$ of [c,b]. Since $S(f; \dot{Q}) = S(f_1; \dot{Q}_1) + S(f_2; \dot{Q}_2)$ and $\|Q_1\| < \delta'$, $\|Q_2\| < \delta''$, Step 1 and the Triangle Inequality give $|S(f; \dot{Q}) - (L_1 + L_2)| \leq |S(f_1; \dot{Q}_1) - L_1| + |S(f_2; \dot{Q}_2) L_2| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon$.
   Direct split when c is a cut point.

4. **Case (ii): c is not a partition point of $\dot{Q}$.** Then $c \in (x_{k-1}, x_k)$ for some $k \leq m$. Define tagged partitions $\dot{Q}_1$ of [a,c] and $\dot{Q}_2$ of [c,b] by $\dot{Q}_1 := \{(I_1, t_1), \ldots, (I_{k-1}, t_{k-1}), ([x_{k-1}, c], c)\}$, $\dot{Q}_2 := \{([c, x_k], c), (I_{k+1}, t_{k+1}), \ldots, (I_m, t_m)\}$. Then all terms of $S(f; \dot{Q})$ and $S(f_1; \dot{Q}_1) + S(f_2; \dot{Q}_2)$ cancel except on I_k, giving $S(f; \dot{Q}) - S(f_1; \dot{Q}_1) - S(f_2; \dot{Q}_2) = (f(t_k) - f(c))(x_k - x_{k-1})$, so $|S(f; \dot{Q}) - S(f_1; \dot{Q}_1) - S(f_2; \dot{Q}_2)| \leq 2M(x_k - x_{k-1}) \leq 2M\|Q\| < 2M \cdot \dfrac{\varepsilon}{6M} = \dfrac{\varepsilon}{3}$.
   Only interval I_k is modified; $|f(t_k) - f(c)| \leq 2M$ and $x_k - x_{k-1} < \varepsilon/(6M)$.

5. Since $\|Q_1\| \leq \|Q\| < \delta'$ and $\|Q_2\| \leq \|Q\| < \delta''$, Step 1 gives $|S(f_1; \dot{Q}_1)L_1| < \varepsilon/3$ and $|S(f_2; \dot{Q}_2) - L_2| < \varepsilon/3$. By the Triangle Inequality, $|S(f; \dot{Q}) - (L_1 + L_2)| \leq |S(f; \dot{Q}) - S(f_1; \dot{Q}_1) - S(f_2; \dot{Q}_2)| + |S(f_1; \dot{Q}_1) - L_1| + |S(f_2; \dot{Q}_2) - L_2| < \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a, b]$ and $\int_a^b f = L_1 + L_2 = \int_a^c f + \int_c^b f$.
   Triangle Inequality and Definition 7.1.1.

($\Rightarrow$)

1. Suppose $f \in \mathcal{R}[a, b]$ and let $\varepsilon > 0$. By the Cauchy Criterion 7.2.1, there exists $\eta_\varepsilon > 0$ such that any tagged partitions $\dot{P}, \dot{Q}$ of [a,b] with norms $< \eta_\varepsilon$ satisfy $|S(f; \dot{P}) - S(f; \dot{Q})| < \varepsilon$.
   Cauchy Criterion 7.2.1($\Rightarrow$).

2 Let $\dot{P}_1, \dot{Q}_1$ be any tagged partitions of [a,c] with $\|P_1\| < \eta_\varepsilon$ and $\|Q_1\| < \eta_\varepsilon$. Adjoin the **same** tagged partition $\dot{R}$ of [c,b] with $\|R\| < \eta_\varepsilon$ to both $\dot{P}_1$ and $\dot{Q}_1$ to obtain tagged partitions $\dot{P} = \dot{P}_1 \cup \dot{R}$ and $\dot{Q} = \dot{Q}_1 \cup \dot{R}$ of [a,b] with $\|P\|, \|Q\| < \eta_\varepsilon$. Then $|S(f_1; \dot{P}_1) - S(f_1; \dot{Q}_1)| = |S(f; \dot{P}) - S(f; \dot{Q})| < \varepsilon$. By the Cauchy Criterion 7.2.1, $f_1 \in \mathcal{R}[a, c]$. By the same argument, $f_2 \in \mathcal{R}[c, b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$ follows from the ($\Leftarrow$) part. ∎

The [c,b] terms $S(f; \dot{R})$ cancel identically; Cauchy Criterion 7.2.1($\Leftarrow$).

Marks: statement 1 · boundedness M>0 & δε = min\(δ',δ'',ε/(6M)\) 2 · Case (i) & Case (ii) estimates 4 · (⇒) Cauchy extension 3.

40 PYQ MTS-25 Q25 also MAT-21 Q37a, MAT-22 Q33

State and prove the Second Form of the Fundamental Theorem of Calculus (Theorems 7.3.5 & 7.3.6); counterexample at a discontinuity.

KNOW FIRST

T23 T24 T22 T17 T07

THEOREM 7.3.5 & THEOREM 7.3.6 · FTC SECOND FORM (BARTLE P. 219)

Theorem 7.3.5: Let $f \in \mathcal{R}[a, b]$ and let f be continuous at a point $c \in [a, b]$. Then the indefinite integral $F(z) := \int_a^z f (z \in [a, b])$ is differentiable at c and F'(c) = f(c).

Theorem 7.3.6: If f is continuous on [a,b], then the indefinite integral $F(z) := \int_a^z f$ is differentiable on [a,b] and F'(x) = f(x) for all $x \in [a, b]$.

SKELETON

For $c \in [a, b)$, given $\varepsilon > 0$ use continuity of f at c to find $\eta_\varepsilon > 0$ so $c \leq x < c + \eta_\varepsilon \Rightarrow f(c) - \varepsilon <$ f(x) $< f(c) + \varepsilon$. For 0 < h < η_ε, write $F(c+h) - F(c) = \int_c^{c+h} f$, sandwich between $(f(c) - \varepsilon)h$ and $(f(c) + \varepsilon)h$, divide by h, and repeat for h < 0.

PROOF OF THEOREMS 7.3.5 AND 7.3.6 (BARTLE PP. 219–220)

1. Suppose first that $c \in [a, b)$ and consider the right-hand derivative of F at c. Since f is continuous at c, given $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $c \leq x < c + \eta_\varepsilon (x \in [a, b])$, then $f(c) - \varepsilon <$ f(x) $< f(c) + \varepsilon$.
   Definition 5.1.1 of continuity of f at c.

2. Let h satisfy 0 < h < η_ε (with $c + h \leq b$). By the Additivity Theorem 7.2.9 (and Corollary 7.2.10), f is integrable on [a,c], [a,c+h], and [c,c+h], and $F(c + h) - F(c) = \int_a^{c+h} f - \int_a^c f = \int_c^{c+h} f$.
   Additivity Theorem 7.2.9 (T17).

3. On the interval [c, c+h], the inequality of Step 1 holds, so Theorem 7.1.5(c) and Example 7.1.4(a) yield $(f(c) - \varepsilon) \cdot h \leq F(c + h) - F(c) = \int_c^{c+h} f \leq (f(c) + \varepsilon) \cdot h$.
   Monotonicity of the integral (Theorem 7.1.5(c), T07) and constant integral $\int_c^{c+h} k =$ kh (T05).

4. Dividing this inequality by h > 0 and subtracting f(c) gives $|\frac{F(c+h) - F(c)}{h} - f(c)| \leq \varepsilon$. Since $\varepsilon > 0$ is arbitrary, the right-hand limit is $\lim_{h \to 0^+} \frac{F(c+h) - F(c)}{h} = f(c)$.
   Definition of right-hand derivative.

5. If $c \in (a, b]$, the left-hand limit is proved identically: for $-\eta_\varepsilon$ < h < 0 (where $f(c) - \varepsilon <$ f(x) < $f(c) + \varepsilon$ on $(c - \eta_\varepsilon, c]$), on [c+h,c] of length -h > 0 we have $(f(c) - \varepsilon)(-h) \leq \int_{c+h}^{c} f = F(c) - F(c+h) \leq (f(c) + \varepsilon)(-h)$; dividing by -h > 0 gives $|\frac{F(c+h) - F(c)}{h} - f(c)| \leq \varepsilon$, so $\lim_{h \to 0^-} \frac{F(c+h) - F(c)}{h} = f(c)$. Hence F is differentiable at c and F'(c) = f(c).
   Agreement of left and right derivatives at interior points (and one-sided derivative at a or b).

6. **Proof of Theorem 7.3.6:** If f is continuous at every point of [a,b], then $f \in \mathcal{R}[a, b]$ by Theorem 7.2.7, and applying Theorem 7.3.5 at each $x \in [a, b]$ shows that F is differentiable on [a,b] with F'(x) = f(x) for all $x \in [a, b]$. ∎
   Pointwise application of Theorem 7.3.5 across [a,b]. (Note: MAT-22 Q33 states Theorem 7.3.5 in full.)

Example where F fails to be differentiable at 0 (Example 7.3.7(a), C20): Let $f(x) := \operatorname{sgn} x$ on [-1,1]. Then $f \in \mathcal{R}[-1, 1]$ (step function), and its indefinite integral with basepoint -1 is $F(x) = \int_{-1}^{x} \operatorname{sgn} = |x| - 1$. At c = 0 (where f is discontinuous), $F'_-(0) = -1 \neq +1 = F'_+(0)$, so F is not differentiable at 0.

Marks: statements of 7.3.5 & 7.3.6 2 · continuity of f at c setup 1.5 · Additivity F(c+h)-F(c) = ∫c(c+h) f 1.5 · sandwiching & division by h 2 · left-hand limit & 7.3.6 1.5 · operatorname(sgn) counterexample 1.5.